

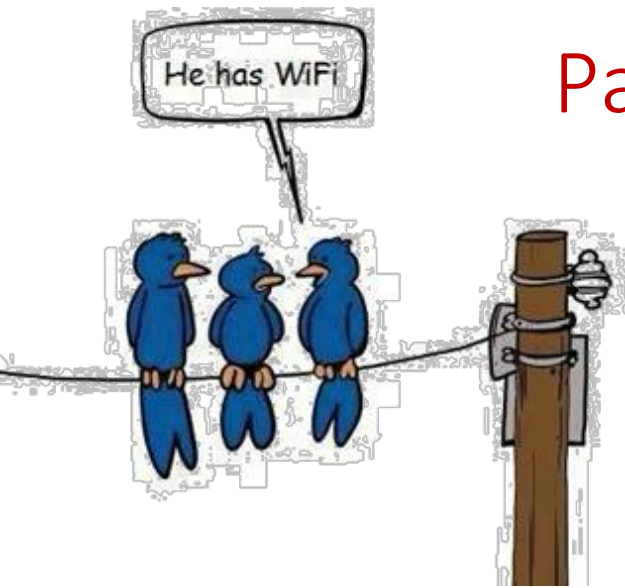
ECSE 4560/ 6560

Modern Communication Systems

Lecture 7: Wireless Systems Overview

Part 6 – Multipath Channel

Ish Jain, Fall 2025



Announcements

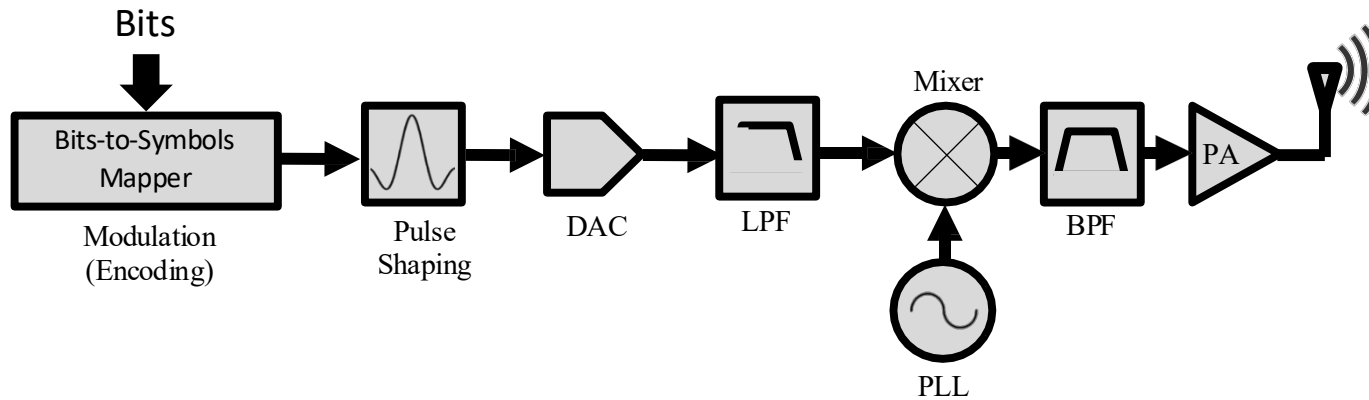
- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Multipath**
- Projects
 - A list of papers will be posted
 - Group formation [MCS Fall25 project.xlsx](#)
 - Proposal due extended **10/3**
- HW1/Lab1 due Monday 9/22 (extended Wed 9/24)
 - Submit on Gradescope – Mark pages for each question parts
 - Matlab coding exercises – “Use Publish option to submit your code”
- HW2/Lab2 posted due Mon 9/29 (extended Wed 10/1)

Today's Class: Wireless Overview 6

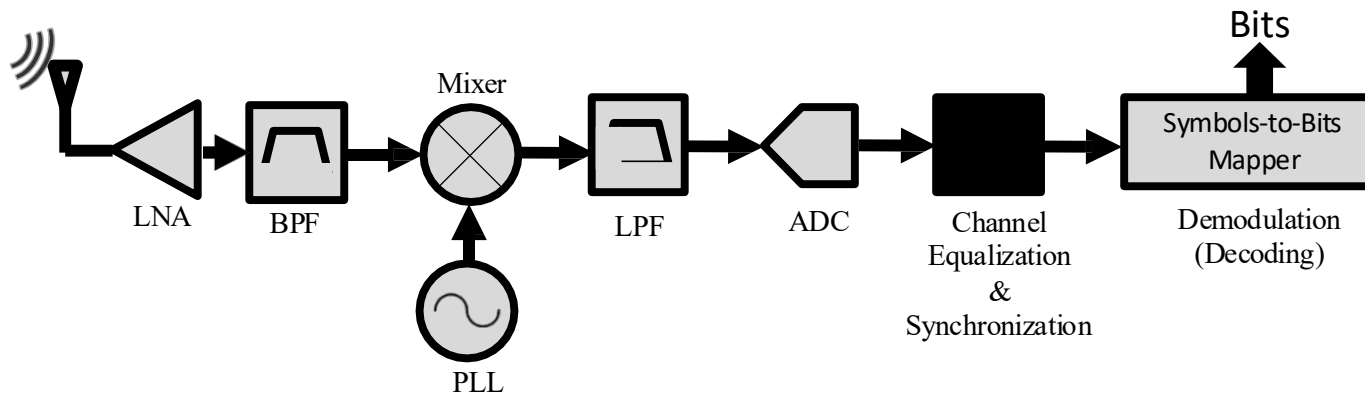
- HW1 and HW2
- Wireless Channel
 - Baseband and passband
- Multipath Channel
 - Power-delay profile
 - Inter-symbol interference
 - Frequency selective fading
 - Narrowband and wideband channel

Wireless Communication System

Transmitter

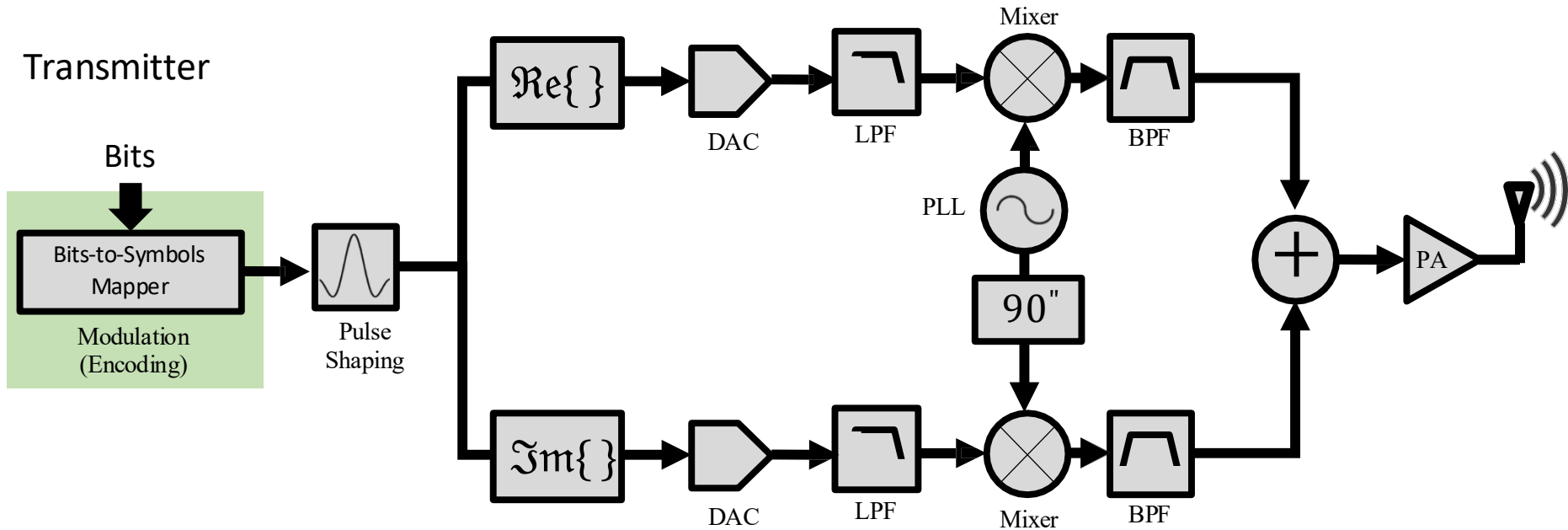


Receiver

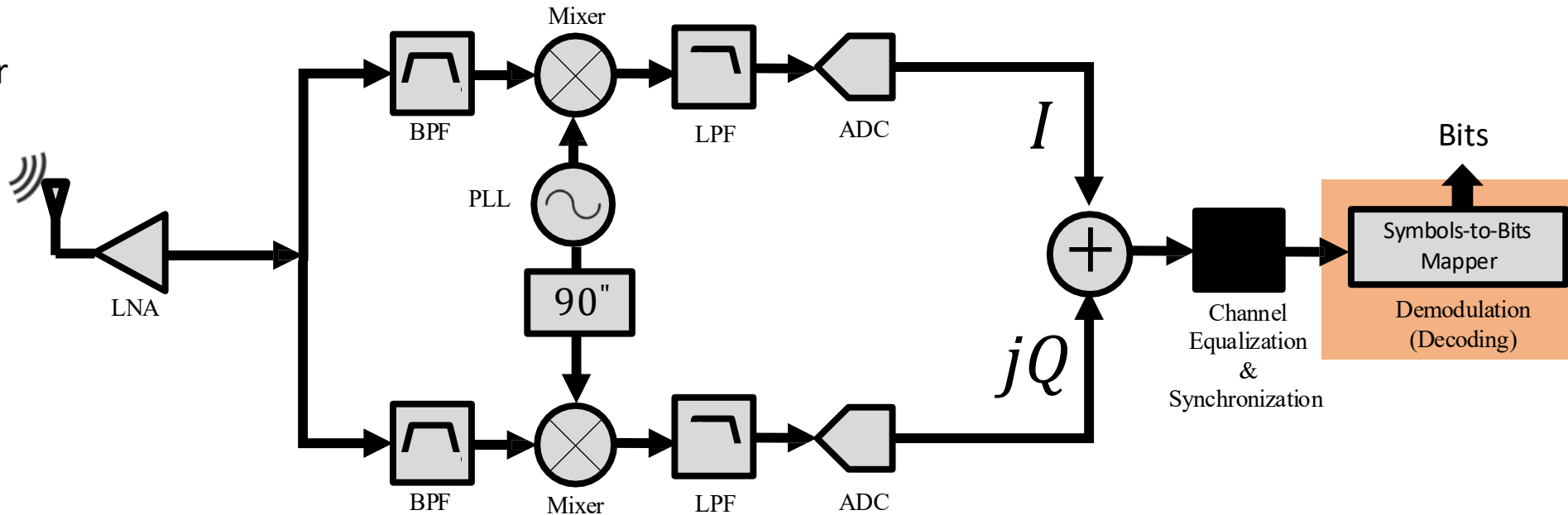


Wireless Communication System

Transmitter



Receiver

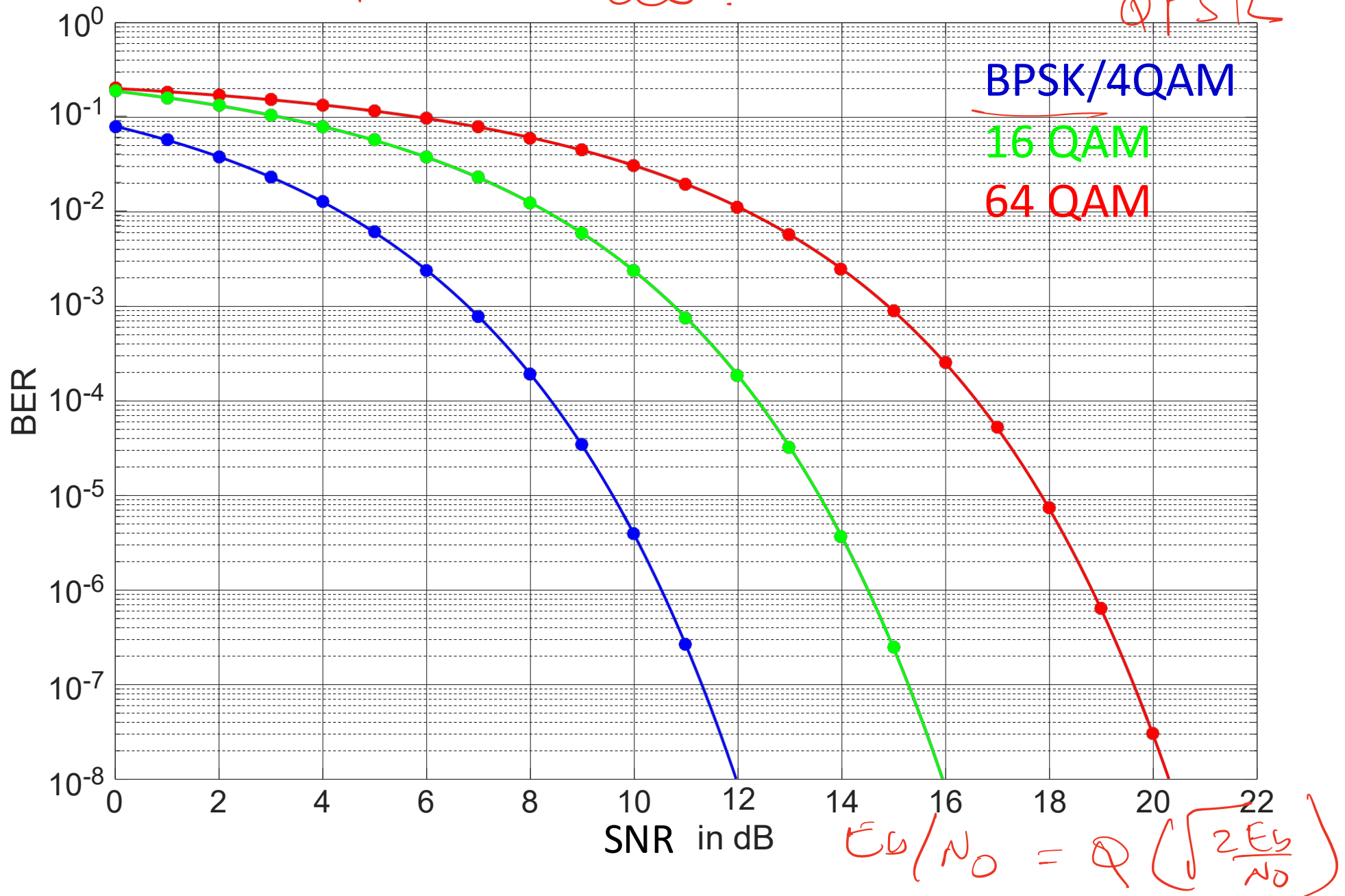


SEER

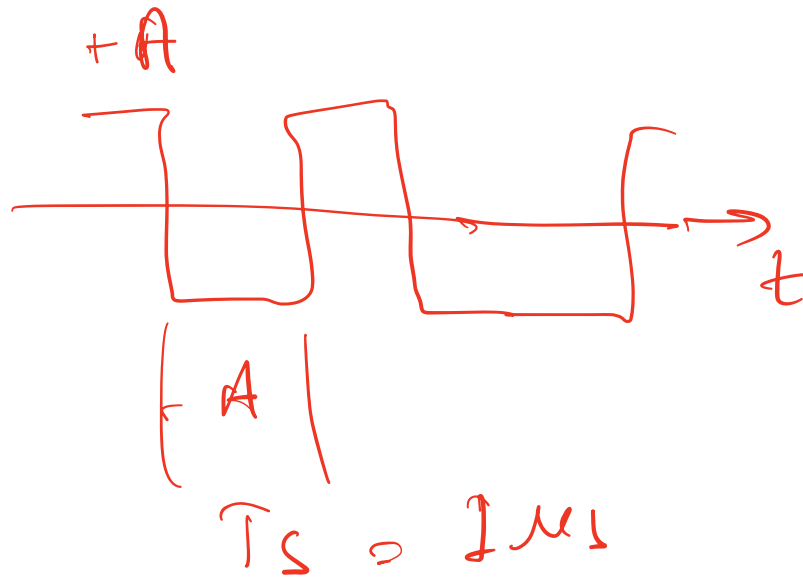
Bit Error Rate (BER)

Monte Carlo.

QPSK

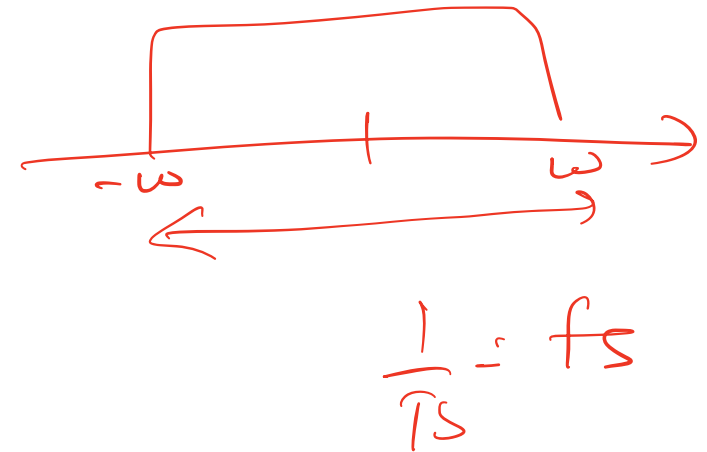


Sampling gain to improve eff. SNR

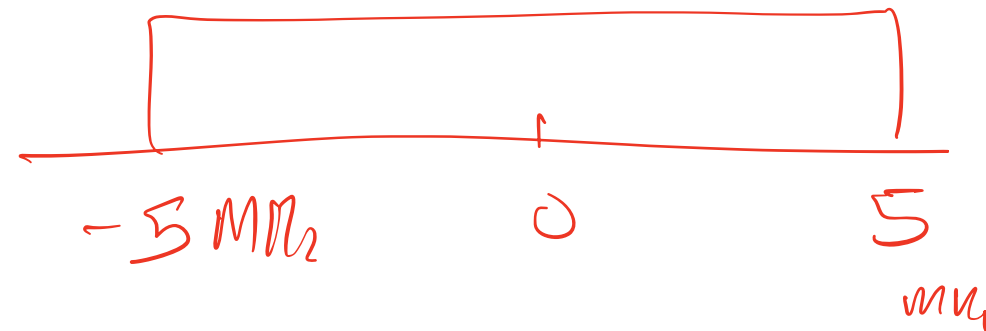


1 Ms/s

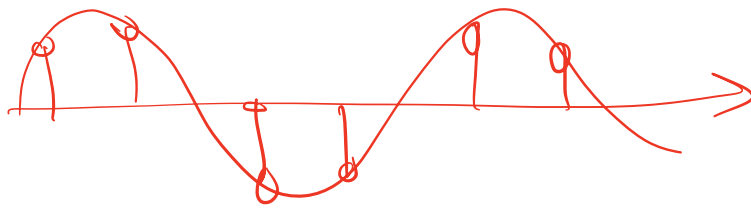
$\downarrow$
 10 Ms/s



$$\omega = \frac{f_s}{2}$$



Processing gain

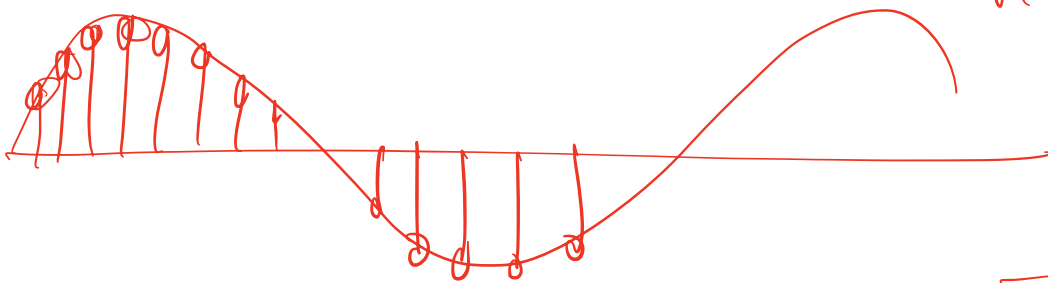


$+A$

$A + n_1$

$A + n_2$

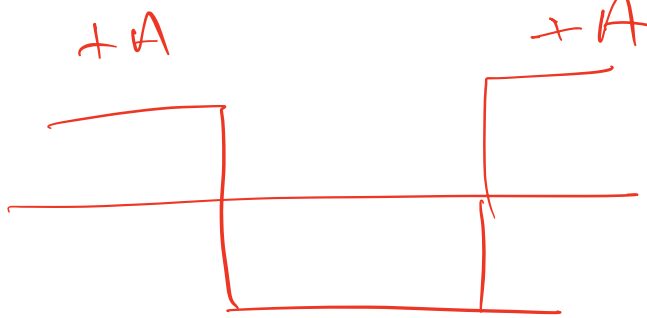
$A + n_3$



Processing gain

$$v \sim N(0, \sigma^2) \quad \frac{1}{3} \sum (A + n_1) + (A + n_2) + (A + n_3)$$

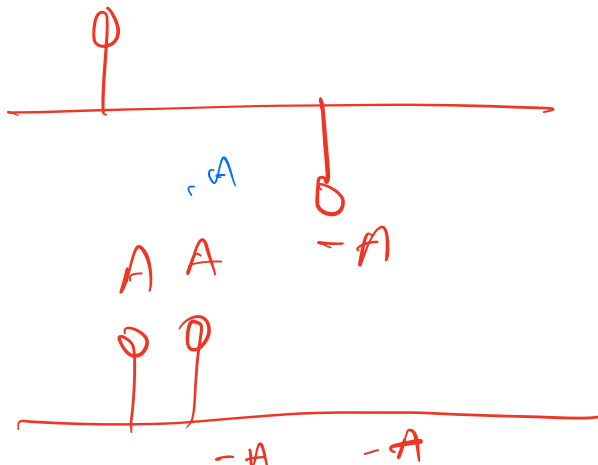
$$= A + \frac{n_1 + n_2 + n_3}{3}$$



$+A$ $-A$

$A + n$

$-A + n$

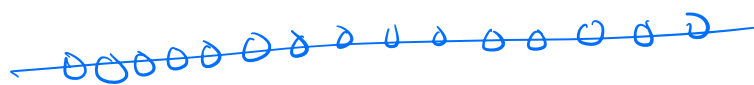
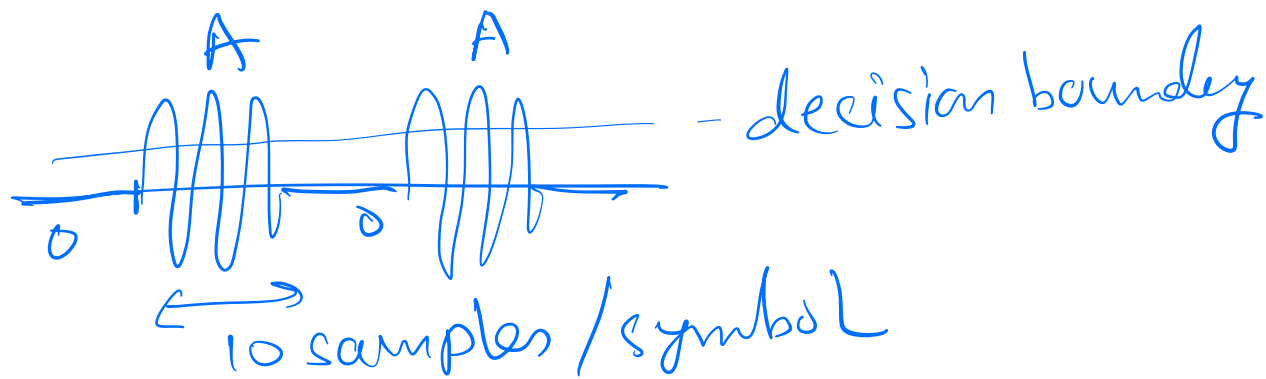
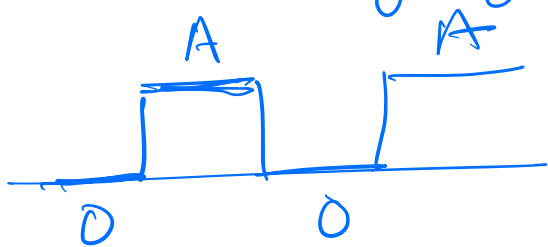


$A + n_1$
 $A + n_2$
 $-A + n_3$
 $-A + n_4$

OOK

Processing gain

0 1 0 1



$$\Rightarrow \frac{1M}{10} = 1e5 \text{ symbols/sec}$$
$$1 \text{ megasamples/sec}$$

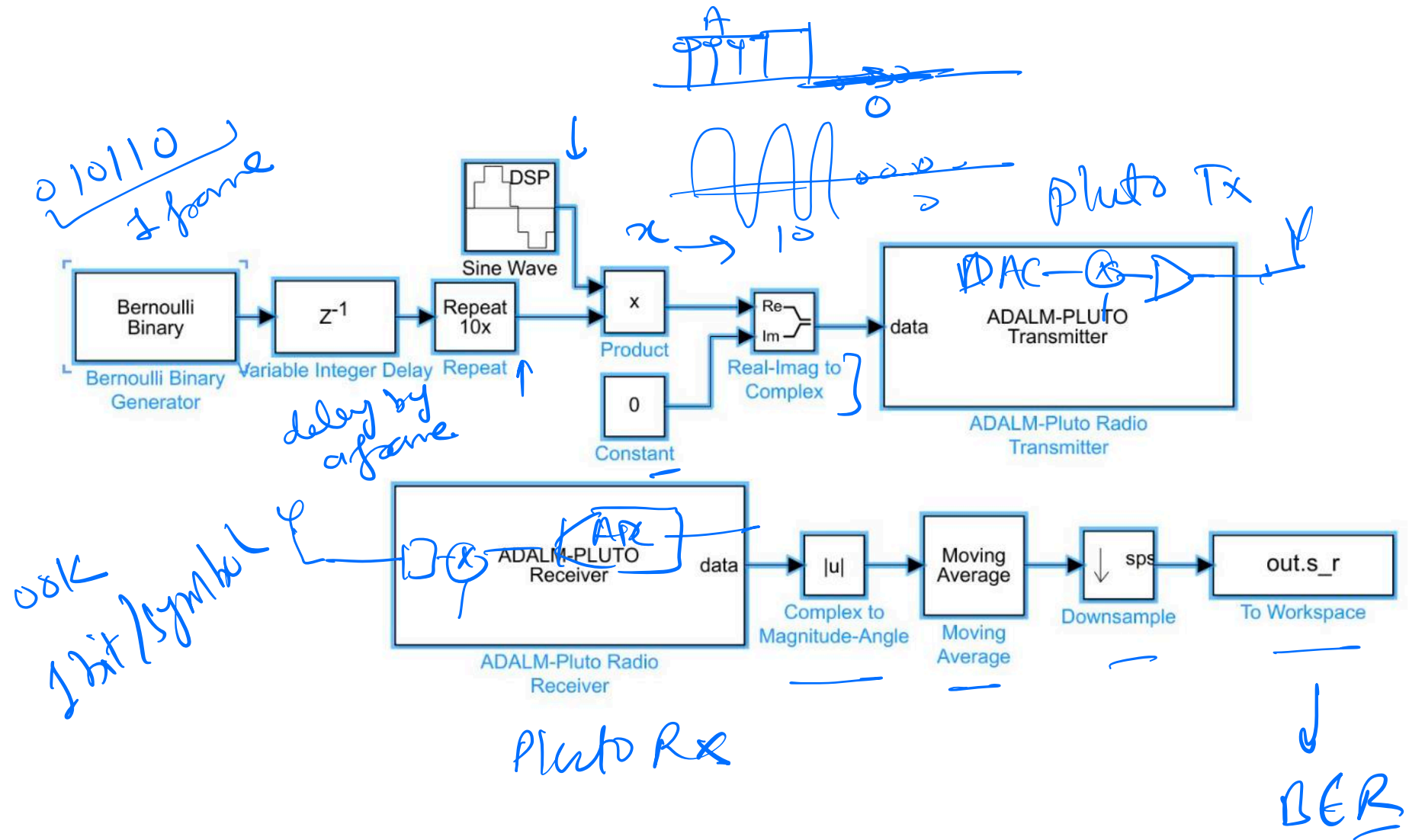
over sampling by 10x

$\Rightarrow$ processing gain

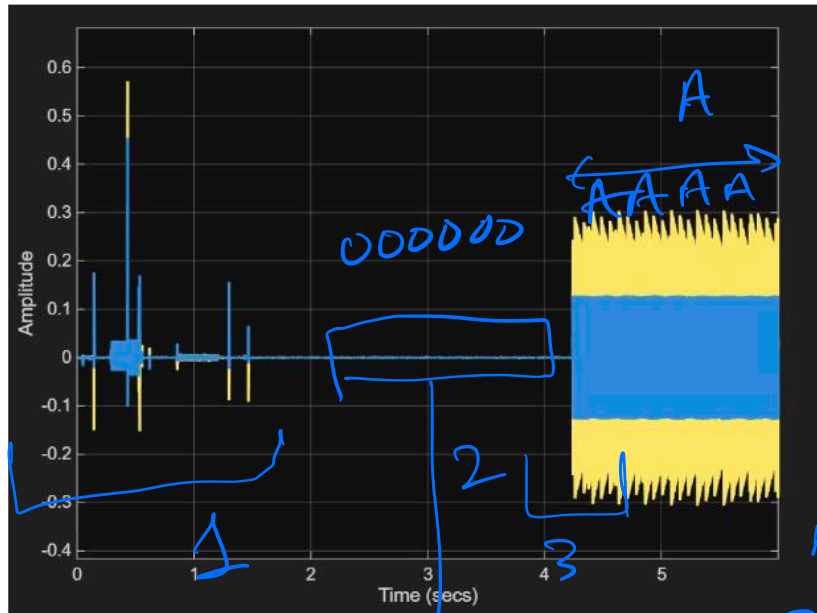
Hardware Pluto+ SDR



Hardware Pluto+ SDR



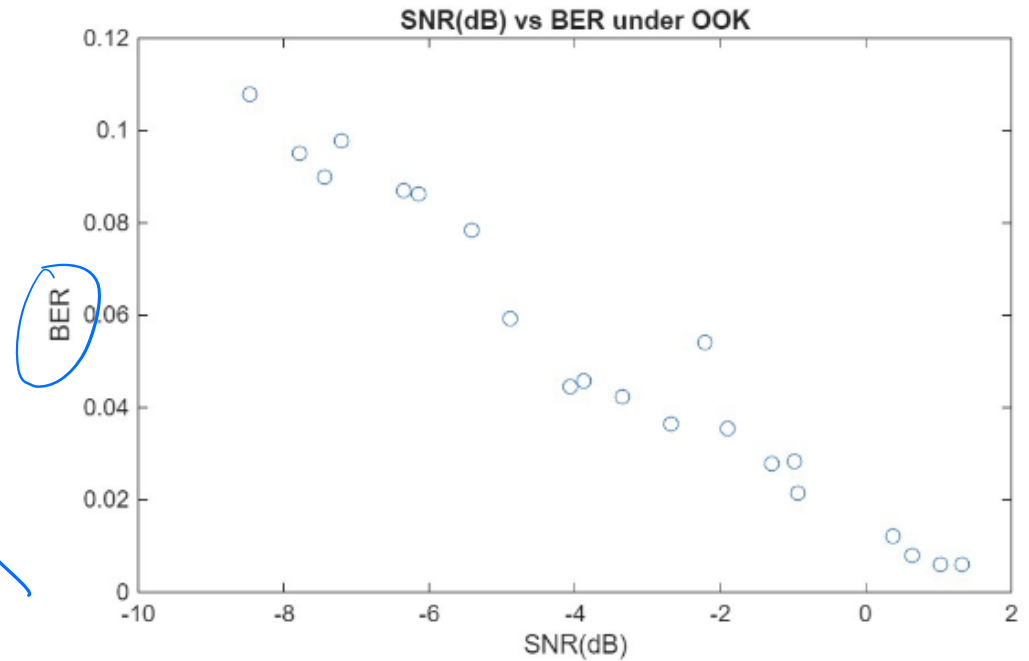
Hardware Pluto+ SDR



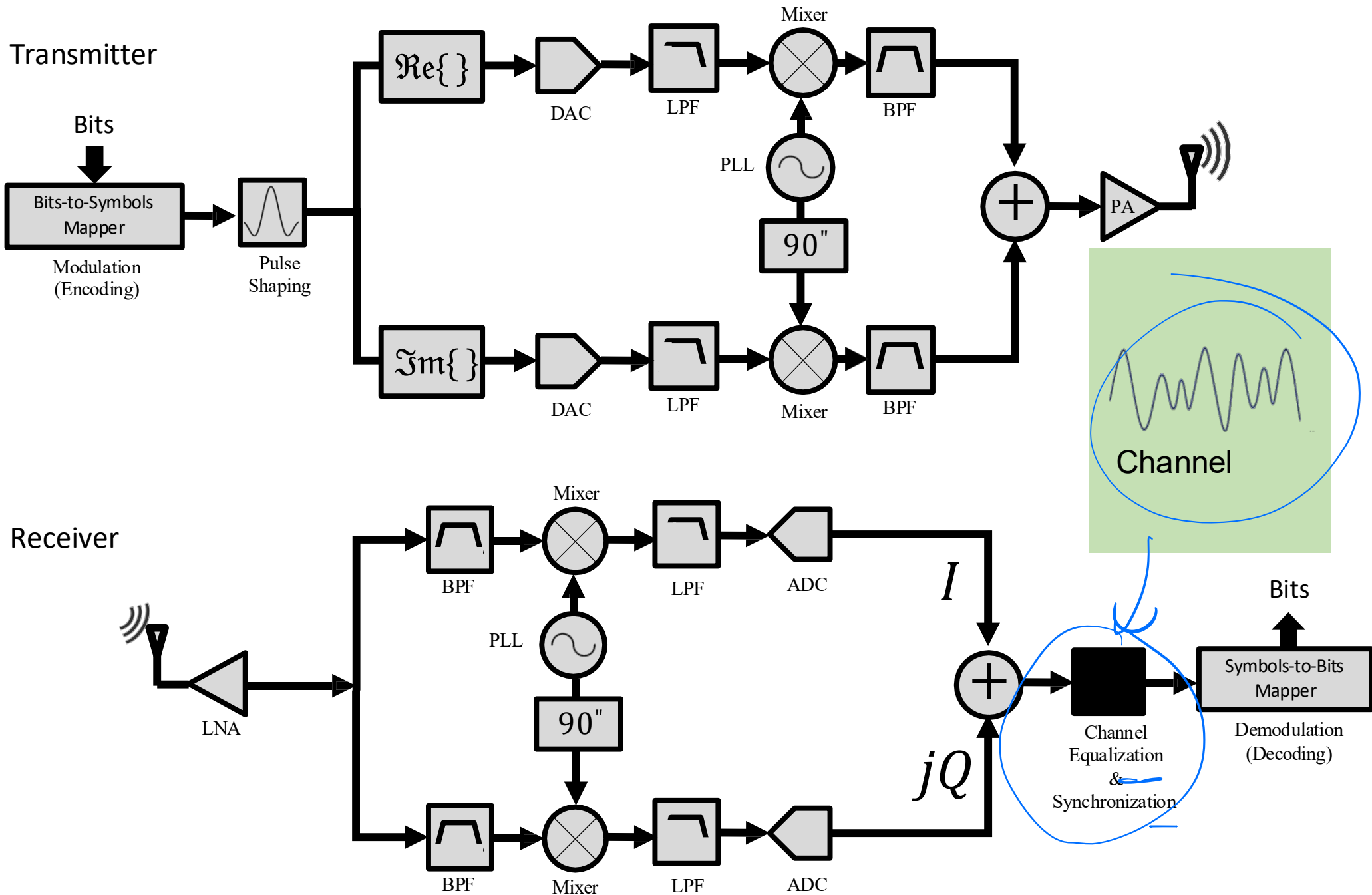
Pluto
glitch.
can be
ignored.

noise

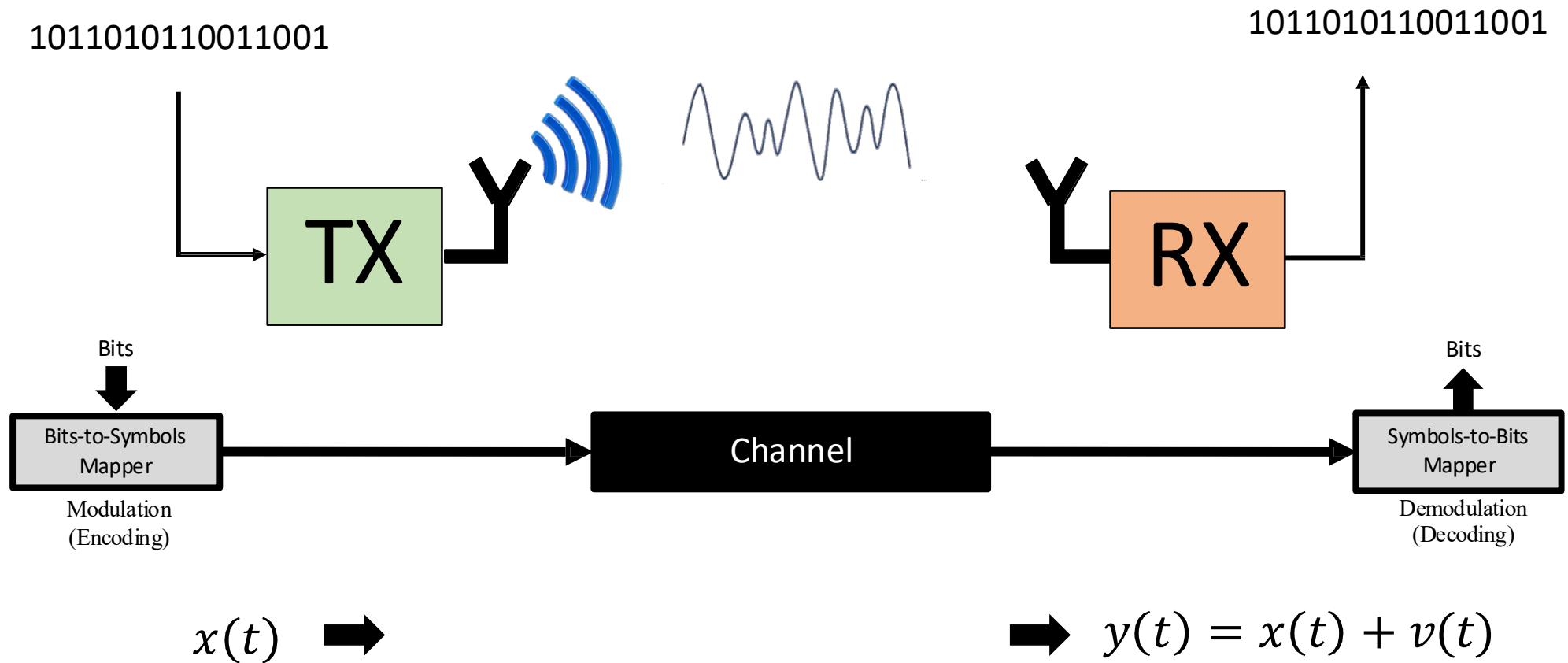
frame
of data



Wireless Communication System



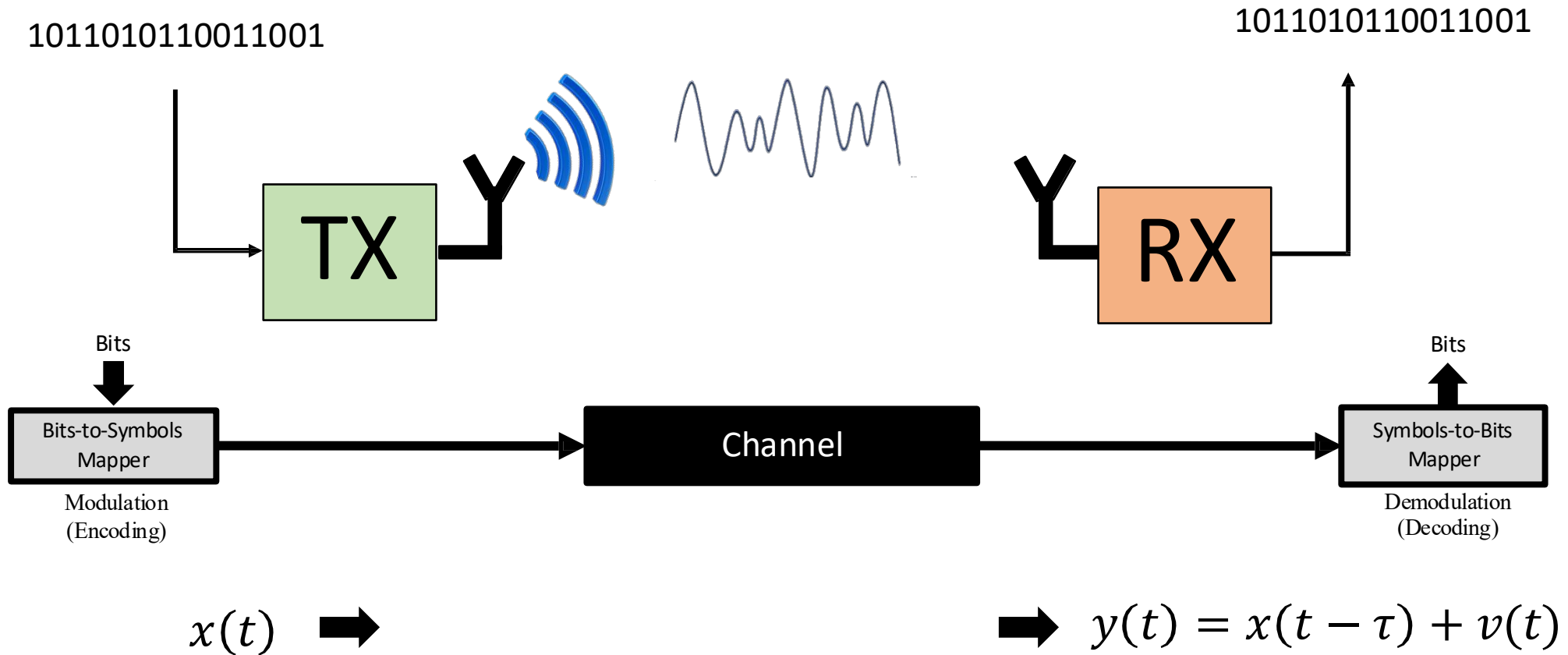
The Channel



Channel adds noise (AWGN)!

$$v(t) \sim N(0, \sigma^2)$$

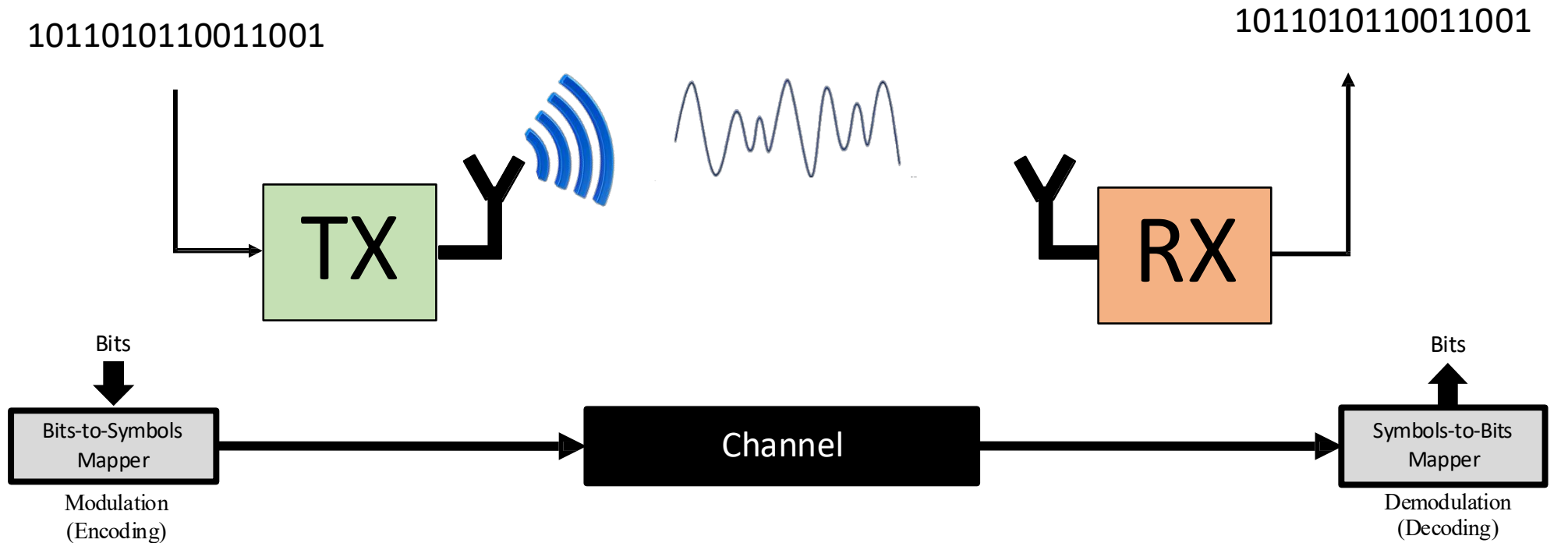
The Channel



Channel delays the signal!

$$\tau = \frac{d}{c}$$

The Channel



$$x(t) \rightarrow$$

$$\rightarrow y(t) = \textcolor{red}{h} x(t - \tau) + v(t)$$

Channel attenuates the signal (Pathloss) *free space*

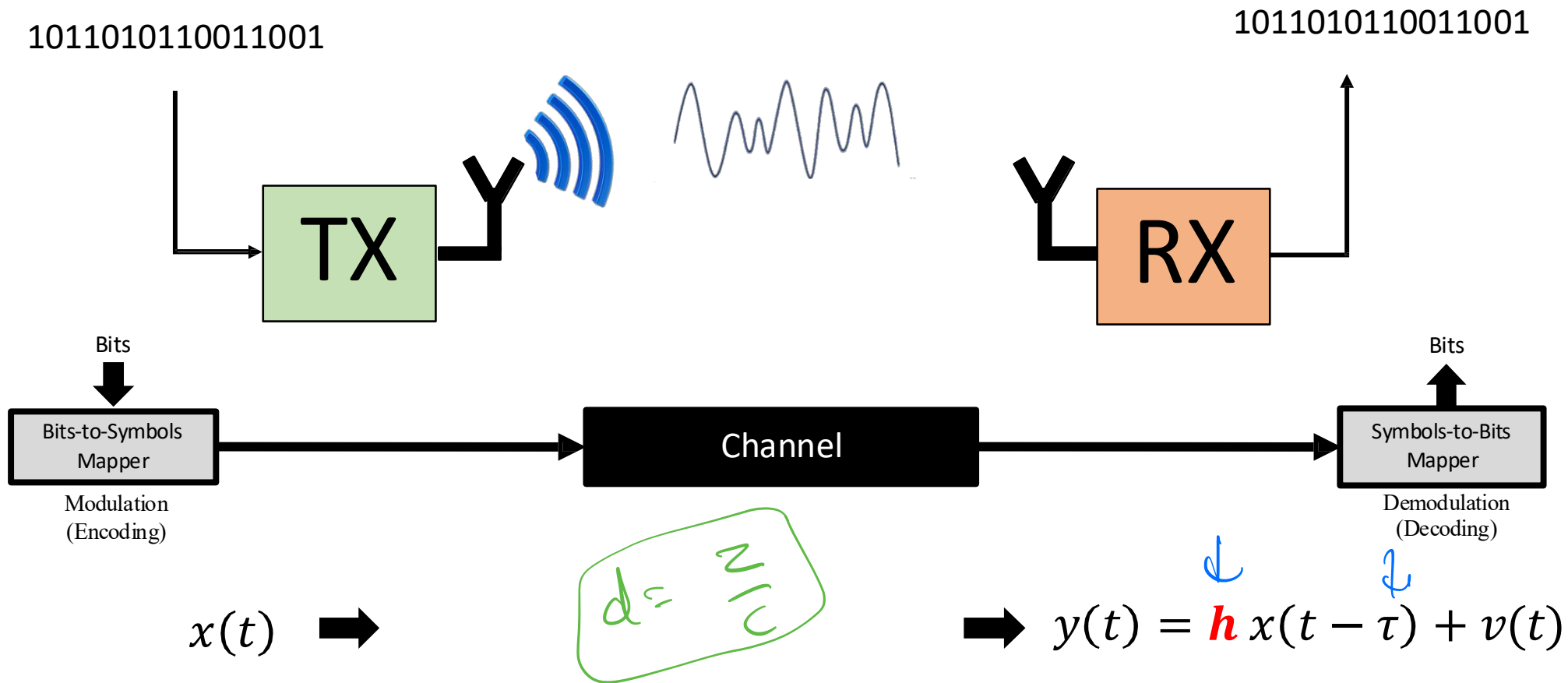
h ∝ 1/d²

γ: 2 for free space

$$P_{RX} = G_{TX} G_{RX} \frac{\lambda^2}{(4\pi d)^2} P_{TX}$$

$$\rightarrow |h| \propto \frac{\lambda}{d}$$

The Channel

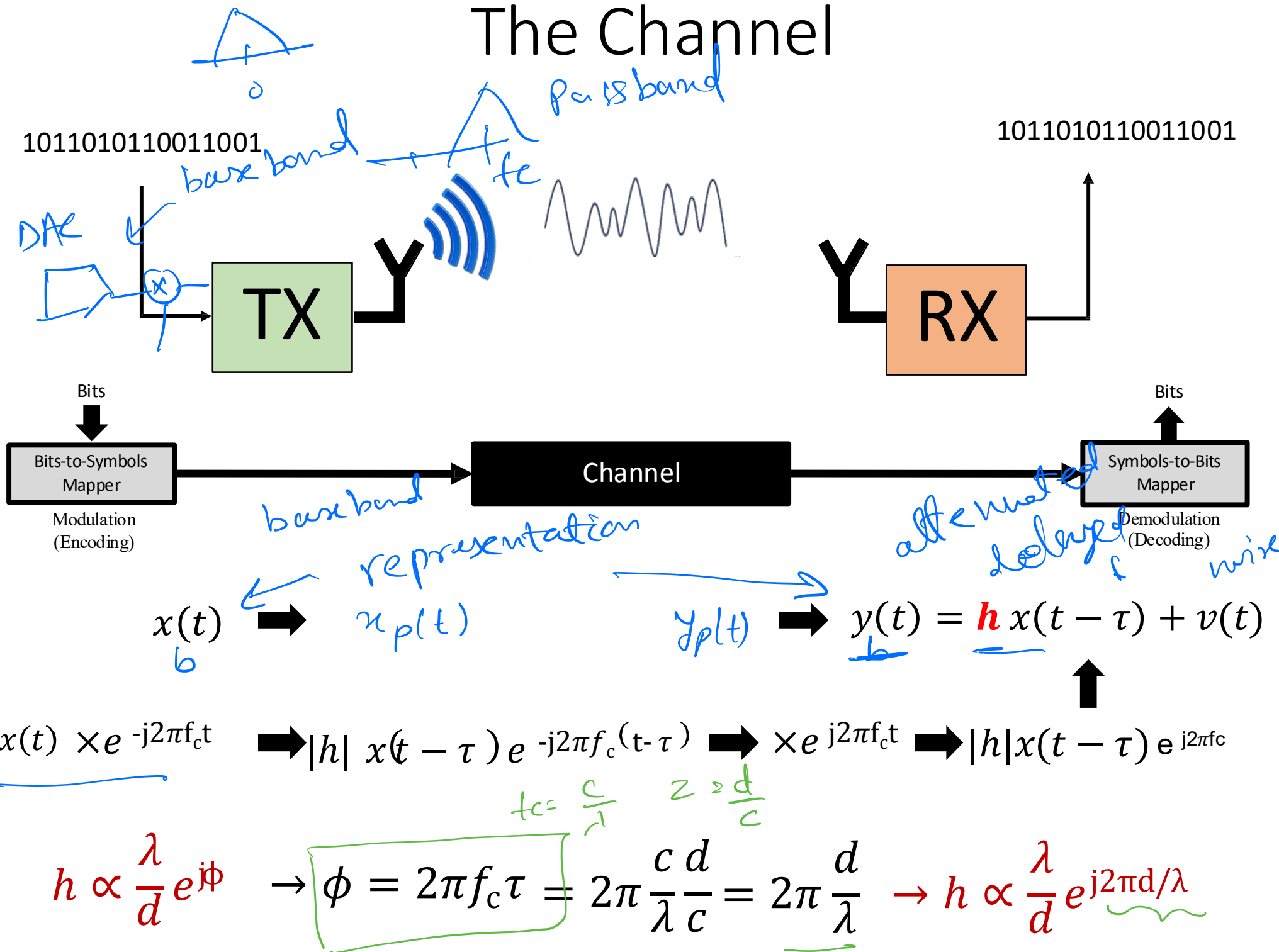


Channel rotates the signal (Adds Phase)

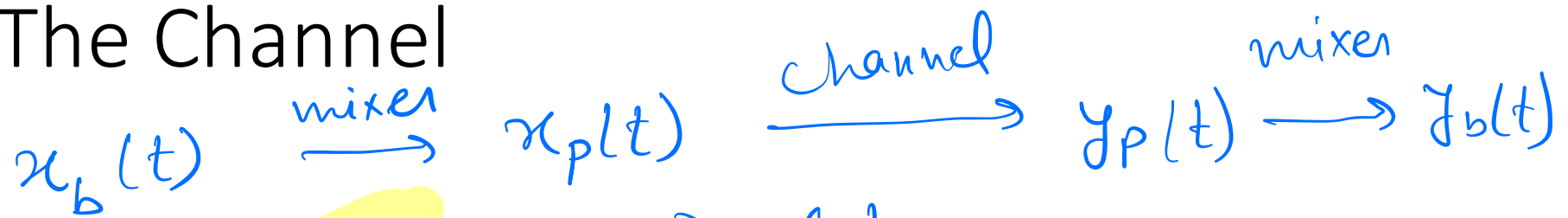
$$h \propto \frac{\lambda}{d} e^{j\phi} \leftarrow \text{artifact of mixer}$$

$f(z, f_c)$

The Channel



The Channel



$$x_p(t) = \text{Re} \{ x_b(t) e^{-j2\pi f_c t} \}$$

$$y_p(t) = h x_p(t - \tau) + v$$

$y_b = f(x_b)$

$$y_b(t) = \frac{y_p(t) e^{j2\pi f_c t}}{e^{j2\pi f_c t}}$$

$$= h x_p(t - \tau) e^{-j2\pi f_c(t - \tau)}$$

$$= \underbrace{h x_b(t - \tau) e^{j2\pi f_c \tau}}_{x_p(t - \tau)} e^{-j2\pi f_c t}$$

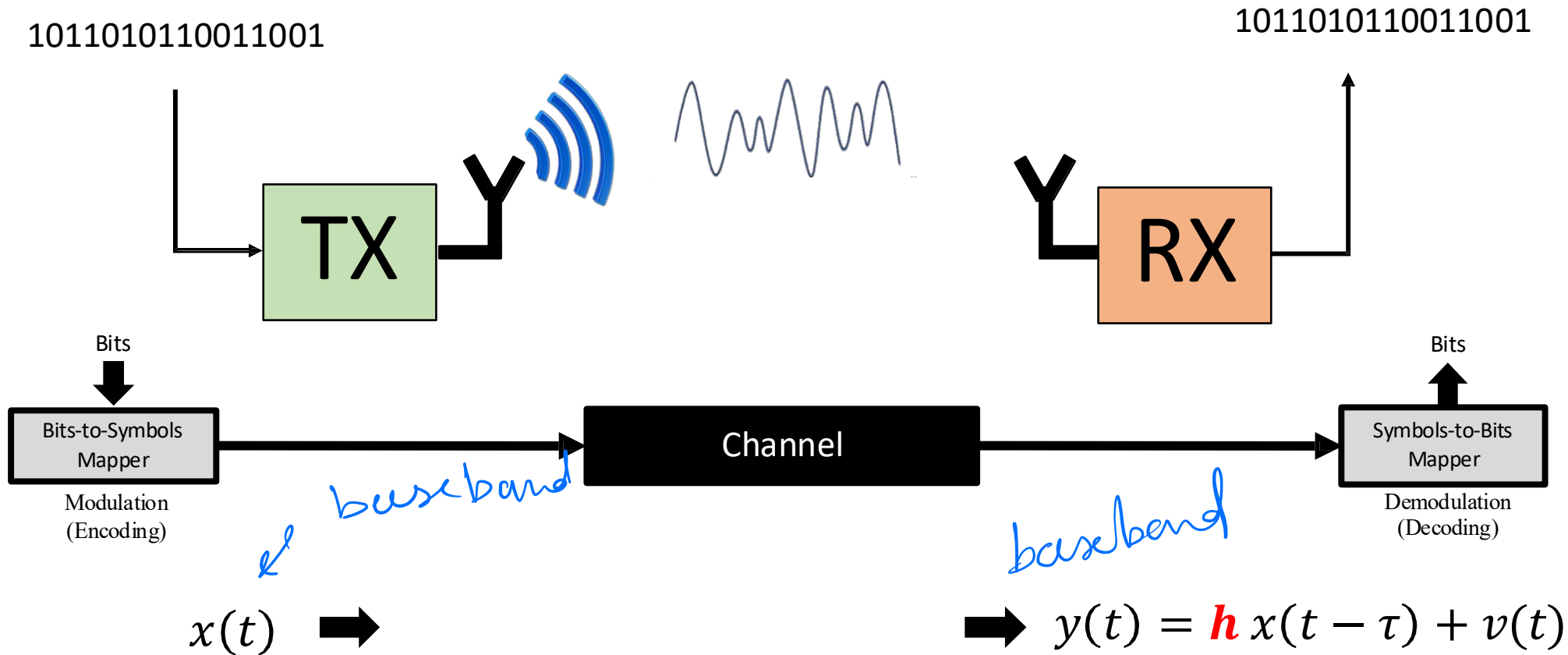
$$= h x_b(t - \tau) e^{-j2\pi f_c t} e^{j2\pi f_c \tau} e^{j2\pi f_c t}$$

$$\underline{y_b(t)} = h e^{j2\pi f_c \tau} x_b(t - \tau) + v_b$$

loss term

Put the value of $y_p(t)$

The Channel



Channel:

- Adds Noise
- Delays the Signal
- Attenuates the Signal
- Rotates the Phase of the Signal

$$h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

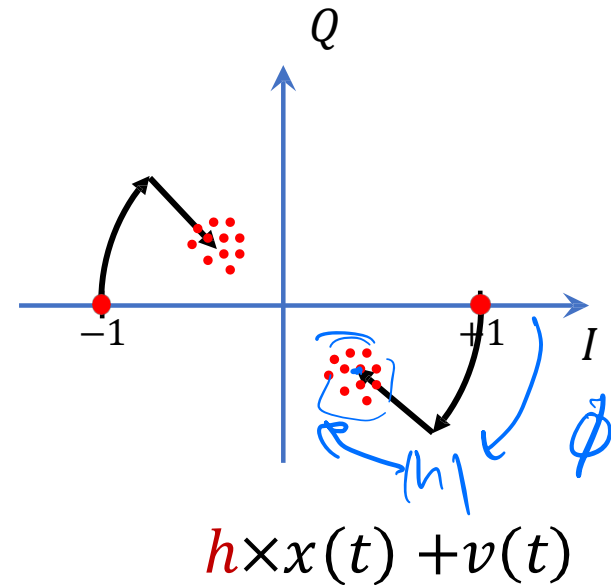
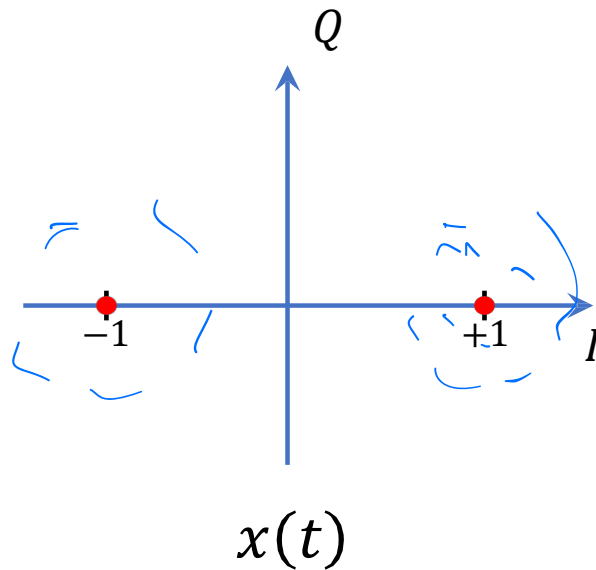
The Channel

Consider BPSK Modulation.

$$0 \rightarrow -1 \quad -A$$

$$1 \rightarrow +1 \quad A$$

$s(t) + \sqrt{2}$



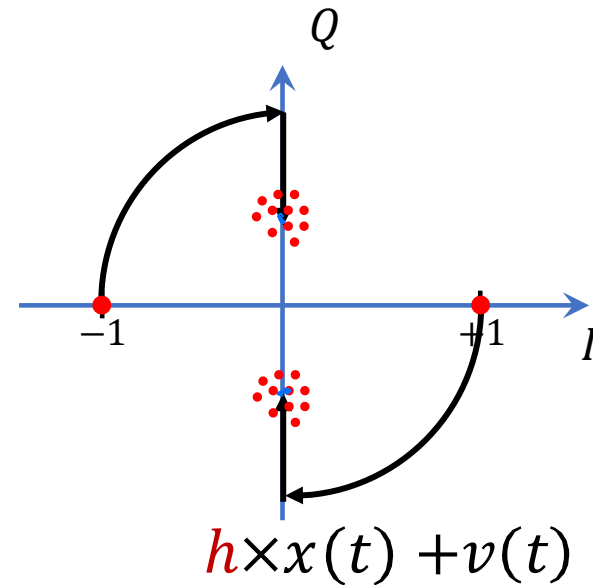
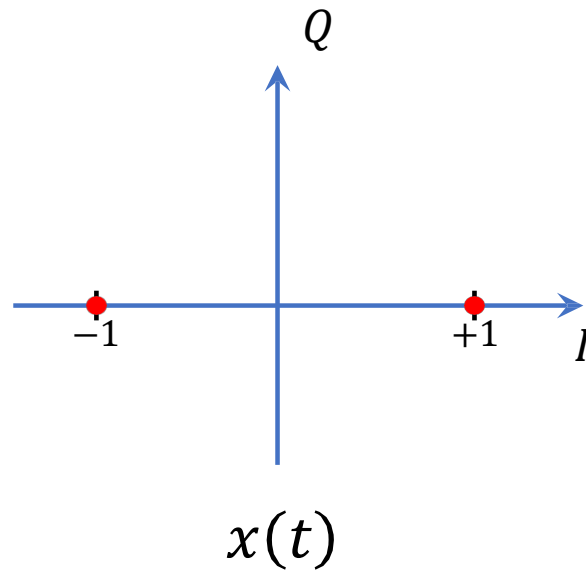
↑ rotation by $\phi = 2\pi d$
attenuation by $|h|$
 $\propto 1/d$

The Channel

Consider BPSK Modulation.

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$

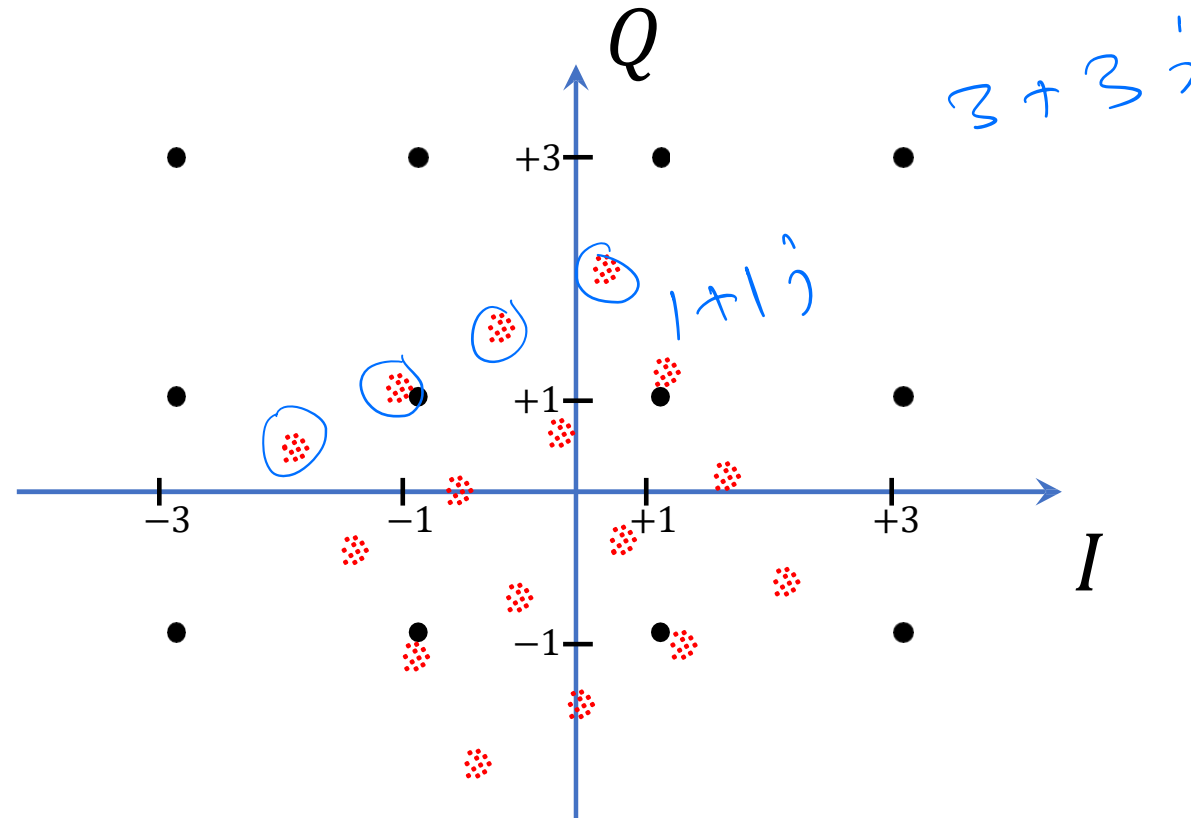


$\varphi > \frac{\pi}{2}$
 $\Rightarrow$ high BER

The Channel

16 QAM

Consider QAM Modulation



Demodulating correctly requires COHERENCE!
i.e., Need to estimate & correct for the channel h



Channel Estimation & Correction

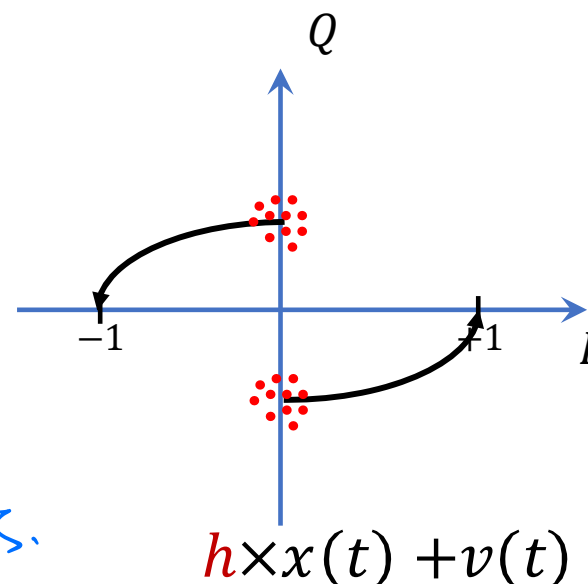
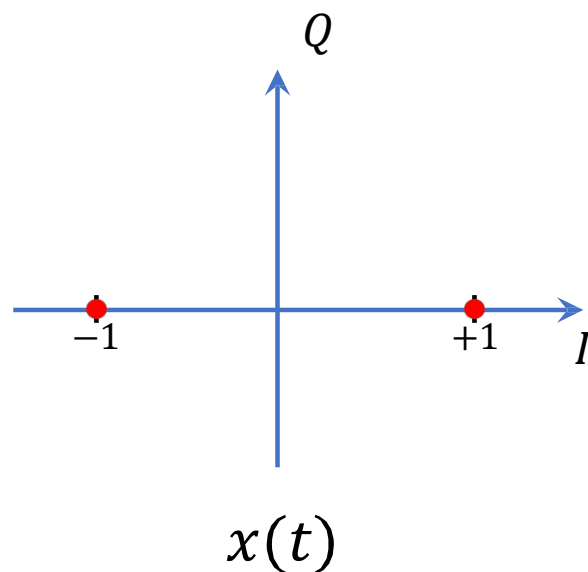
Channel Estimation & Correction

Consider BPSK Modulation.

MLE

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$



Send Training Sequence (Preamble Bits): Known Bits

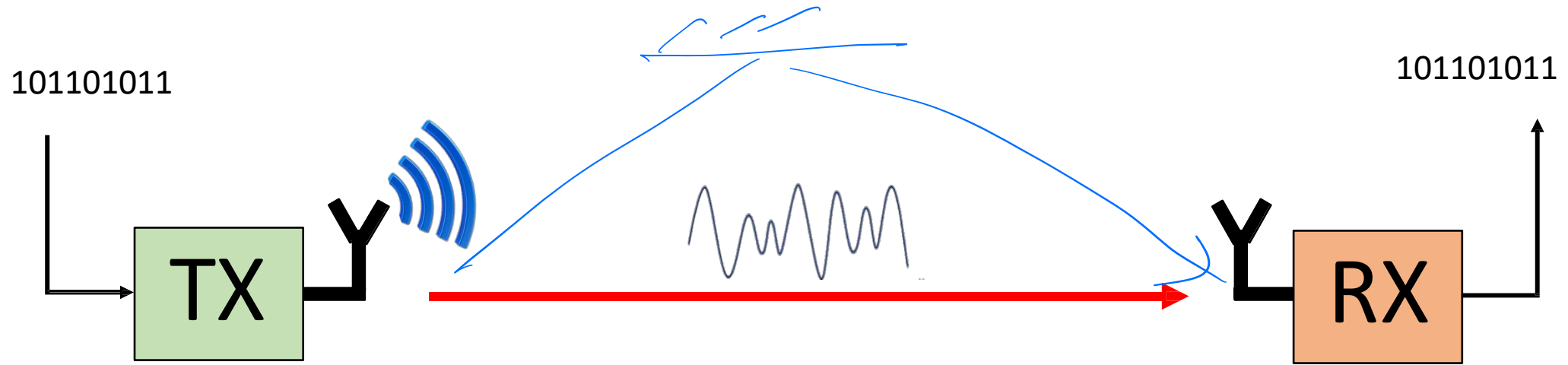
$$\begin{aligned} x(0) = 1 & \longrightarrow y(0) = h + v(0) \\ x(1) = 1 & \longrightarrow y(1) = h + v(1) \\ x(2) = -1 & \longrightarrow y(2) = -h + v(2) \end{aligned}$$

•
•
•

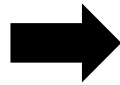
Estimate channel: $\tilde{h} = \frac{1}{K} \sum_{k=1}^K \frac{y(k)}{x(k)}$ ← Known Pilots

Correct channel: $\tilde{x}(t) = \frac{y(t)}{\tilde{h}(t)}$

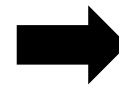
The Channel



$x(t)$



Channel



$y(t) = \mathbf{h} x(t - \tau) + v(t)$

$$h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

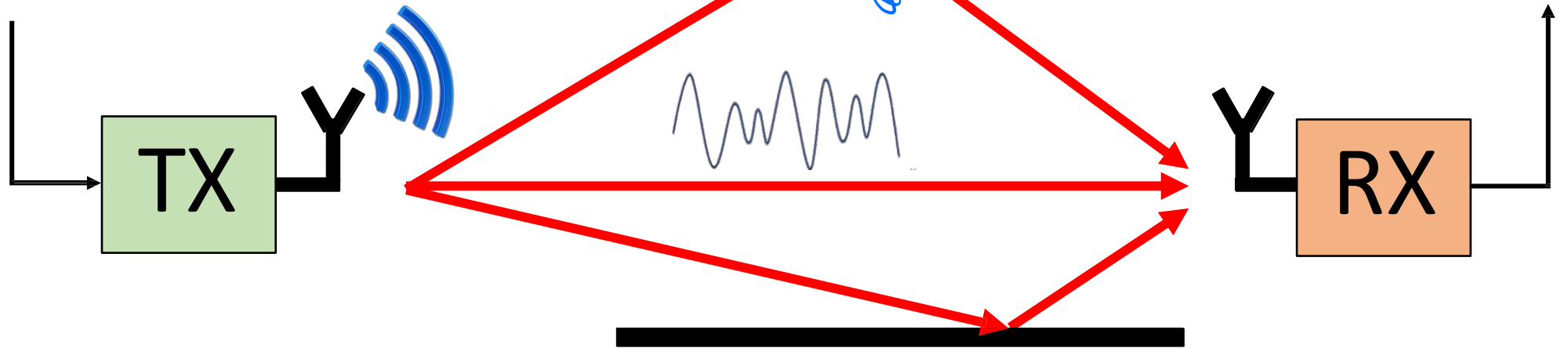
Assumes single path!



Multipath Channel

Reflection
=> Dominant

101101011



Multipath Propagation: radio signal reflects off objects ground, arriving at destination at slightly different times

$$y(t) = \alpha_1 e^{j\phi_1} x(t - \tau_1) + \alpha_2 e^{j\phi_2} x(t - \tau_2) + \alpha_3 e^{j\phi_3} x(t - \tau_3) \dots$$

Handwritten annotations: 'atten.' under α_k and 'delay' under τ_k , with arrows pointing to the respective terms in the equation.

$$y(t) = \sum_k \alpha_k e^{j\phi_k} x(t - \tau_k) = \sum_k h_k(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Handwritten annotations: A bracket under the summation index k and a bracket under the summation term $h_k(\tau_k)$.

$h(t)$ is channel impulse response.

$$h(t) = \sum_k \alpha_k e^{j\phi_k} \delta(t - \tau_k)$$

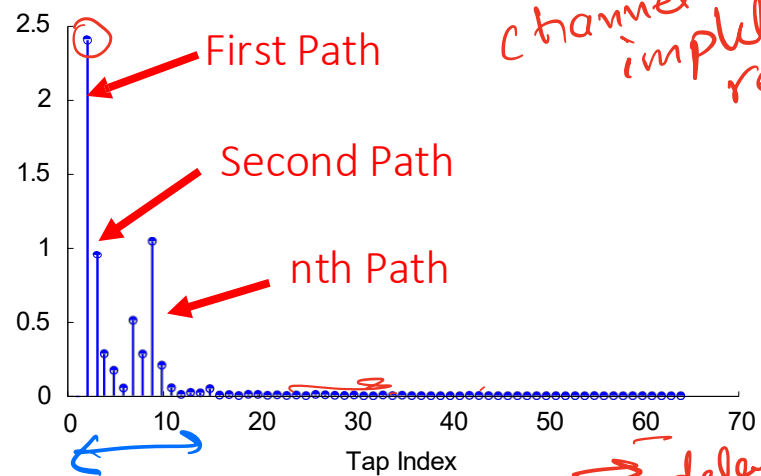
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

n LOS non-line of sight

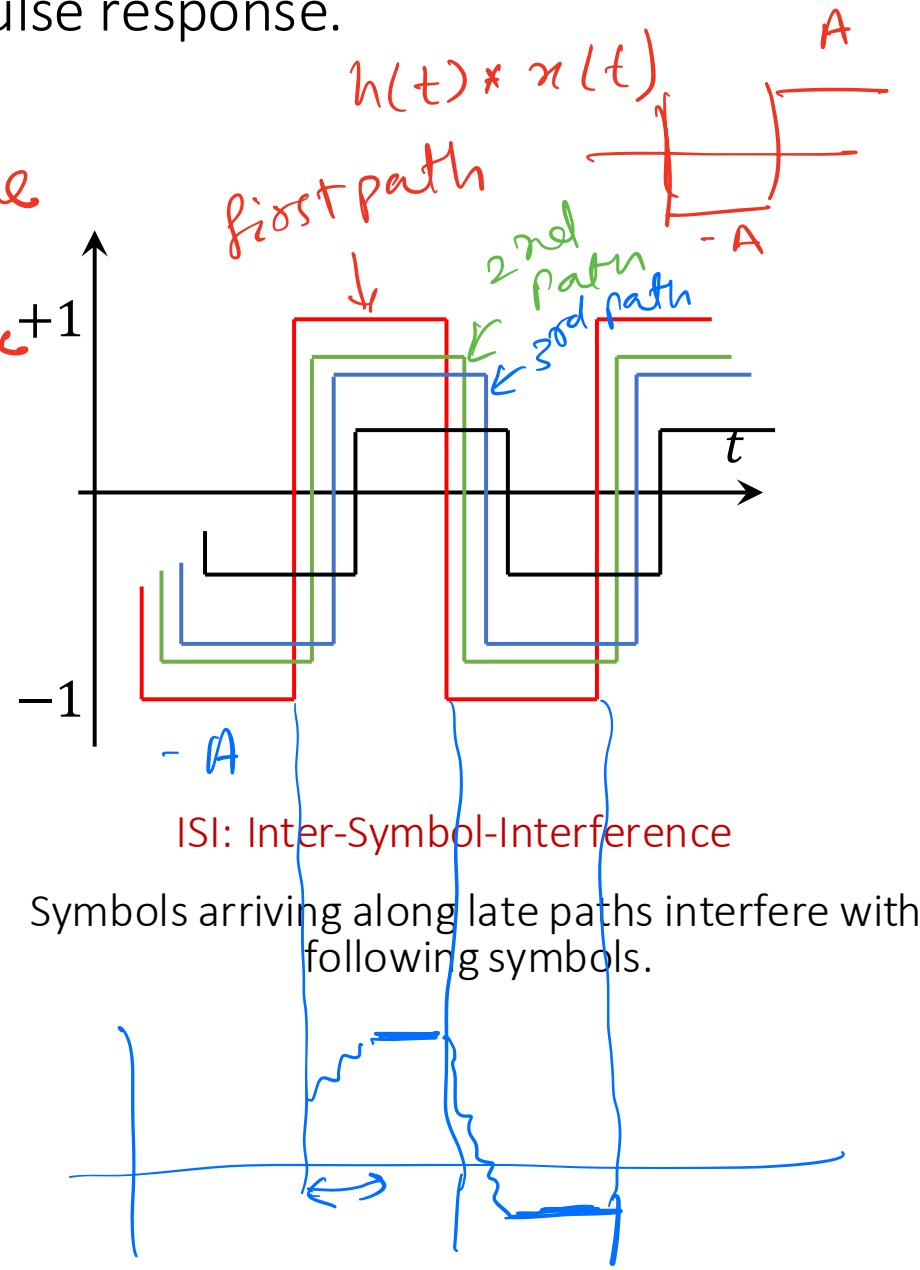
$|h|^2$



*Power-delay profile
channel impulse response*

delay index

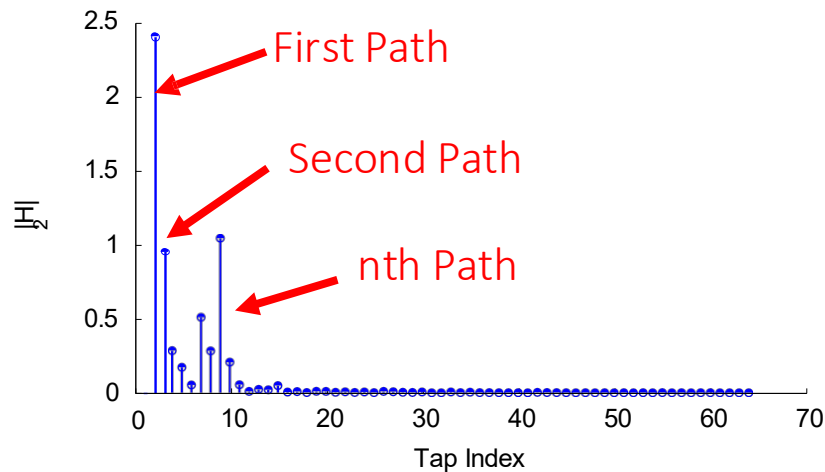
Multi-tap Channel



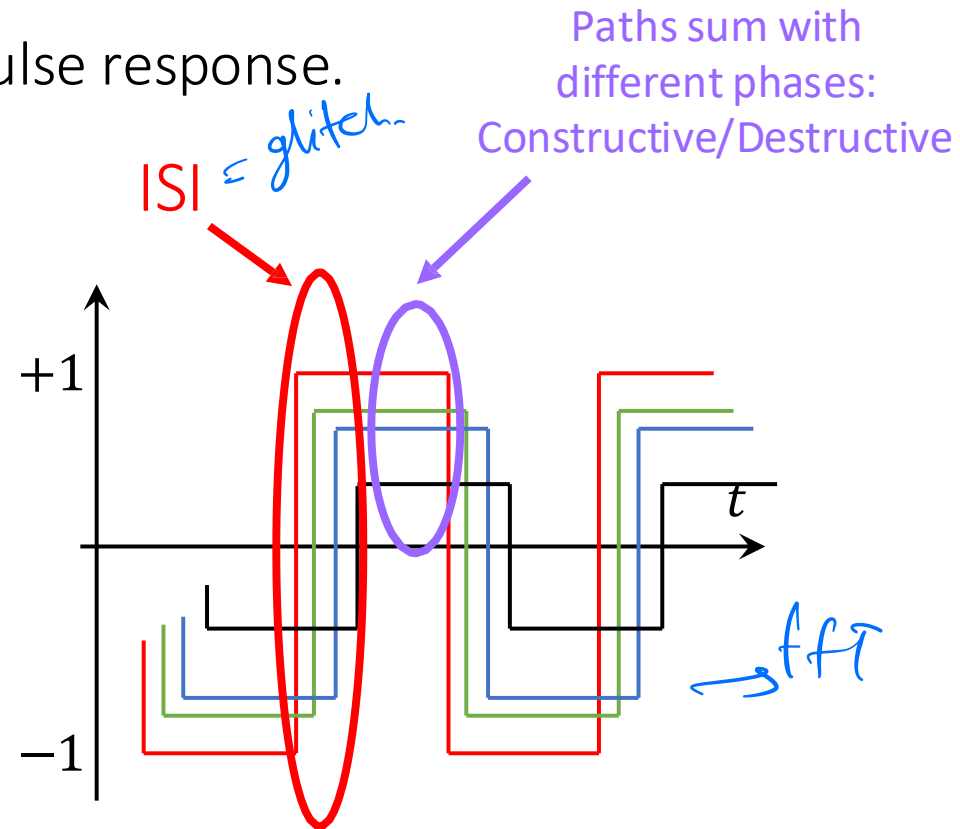
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel



Symbols arriving along late paths interfere with following symbols.



Channel Fading

Symbols arriving along different paths sum up destructively

Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m$, $d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

$$= \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} \left(1 + \frac{d_1}{d_2} e^{j2\pi(d_2-d_1)/\lambda} \right) \quad \frac{d_1}{d_2} \approx 1$$

$$\text{if } \frac{d_2 - d_1}{\lambda} \approx \frac{1}{2} \rightarrow h = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} (1 + e^{j\pi}) = 0 \quad \text{Destructive Interference!}$$

Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

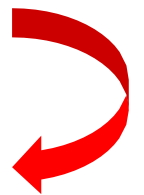
Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m, d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

@ $f_1 = 2.5GHz$ ($\lambda = 12\text{ cm}$): $h = 0.12 e^{j\frac{2\pi}{3}} + 0.113 e^{j\frac{5\pi}{3}} \approx 0.006$

@ $f_2 = 5GHz$ ($\lambda = 6\text{ cm}$): $h = 0.06 e^{j\frac{5\pi}{3}} + 0.05 e^{j\frac{5\pi}{3}} \approx 0.116$



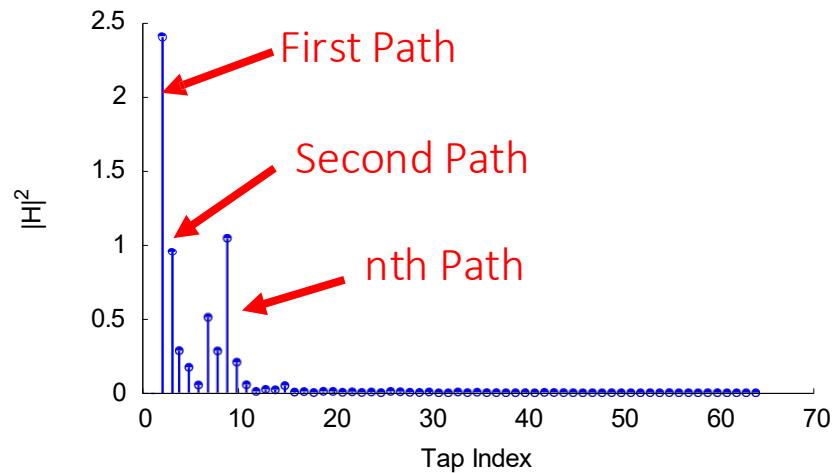
17×
(24dB)

Frequency Selective Fading

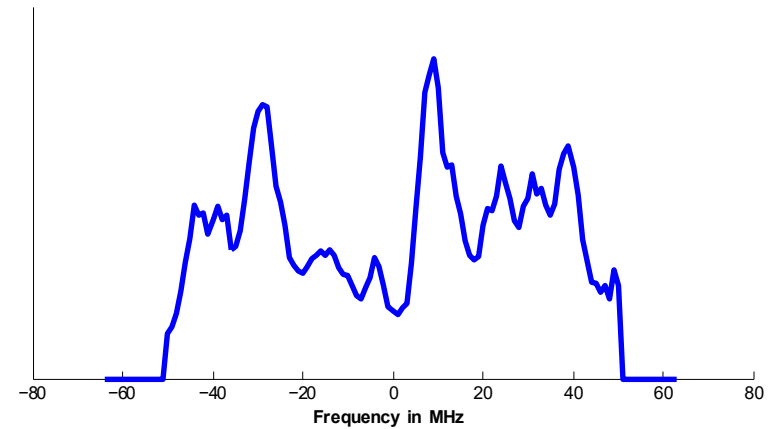
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



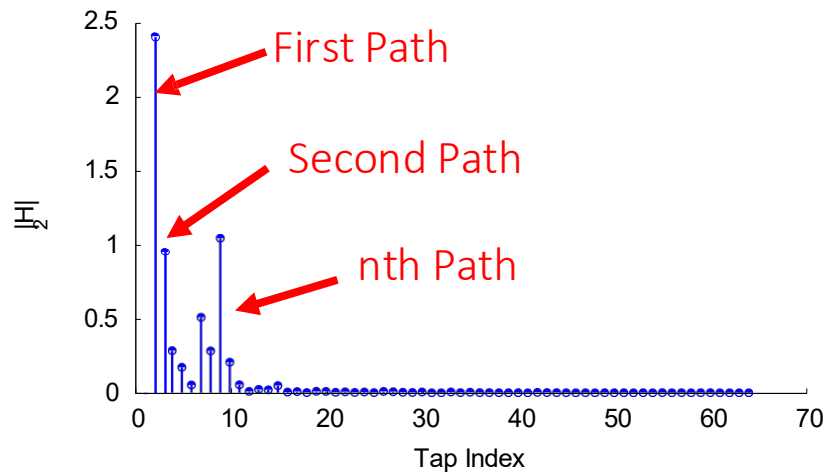
Multi-tap Channel



Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$

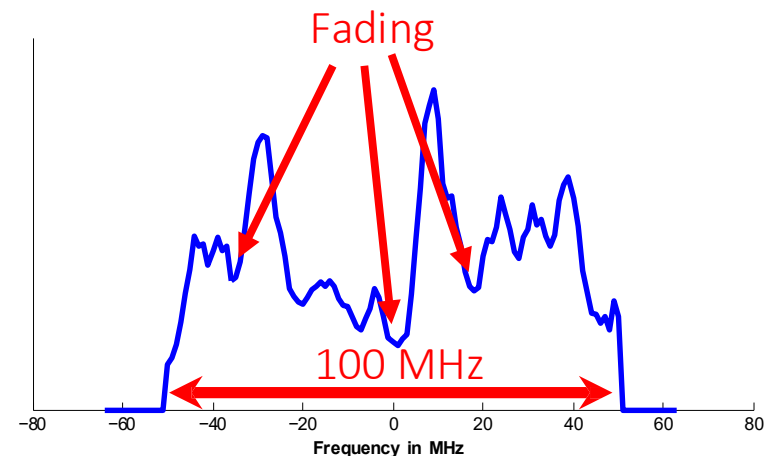


Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.



Frequency Selective Fading

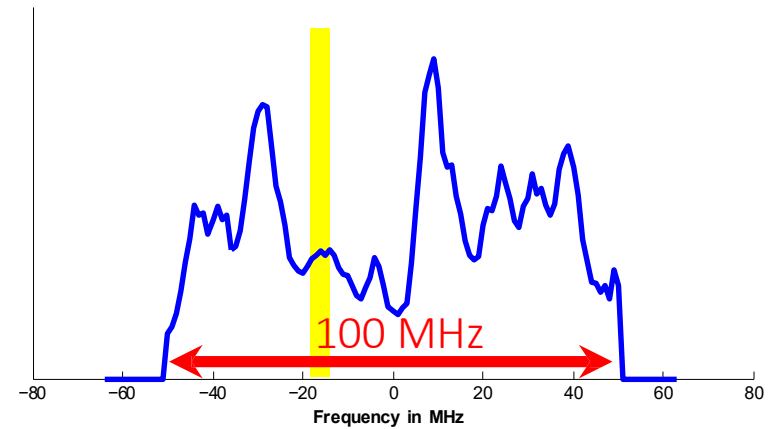
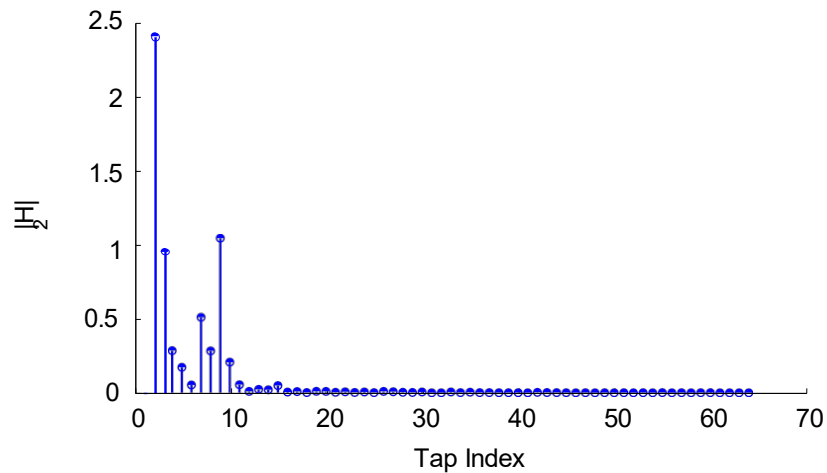
Symbols arriving along different paths sum up destructively

Problematic in
Wideband Channel!

Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$

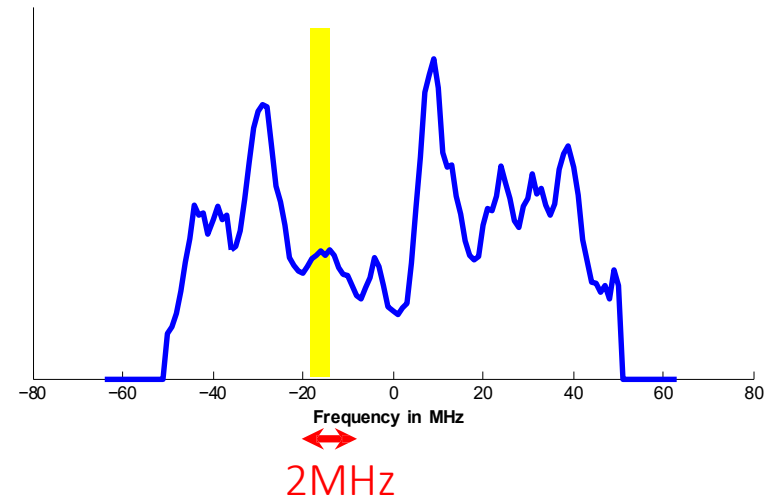
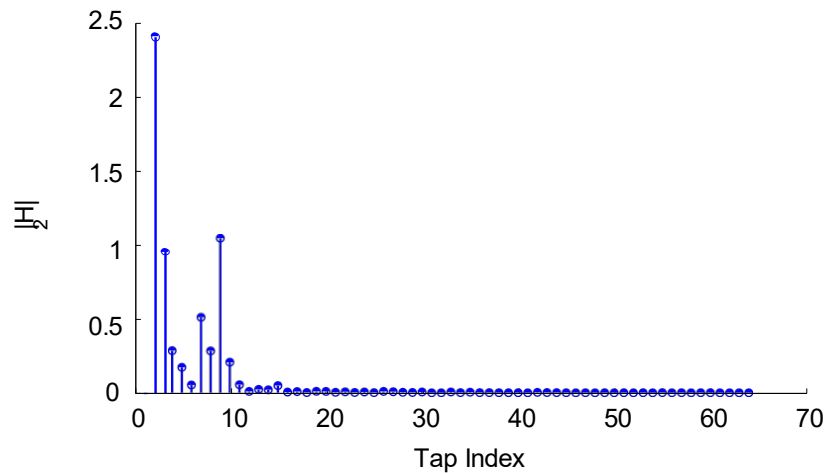


↓
OFDM

Narrowband Channel

$h(t)$ is channel impulse response.

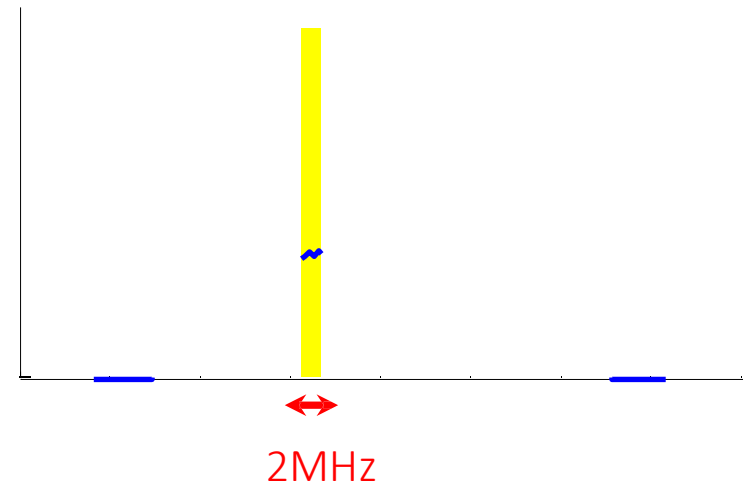
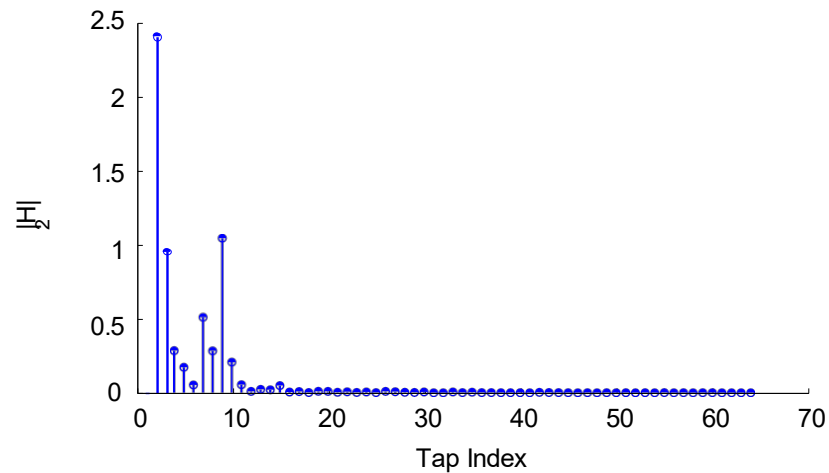
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Narrowband Channel

$h(t)$ is channel impulse response.

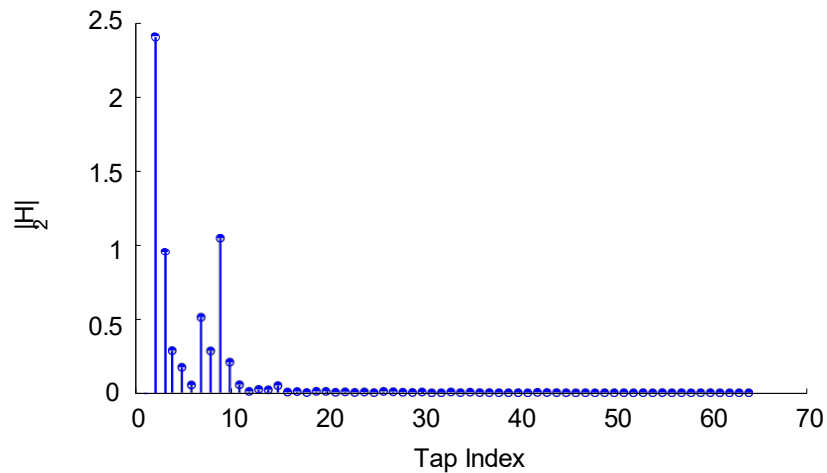
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



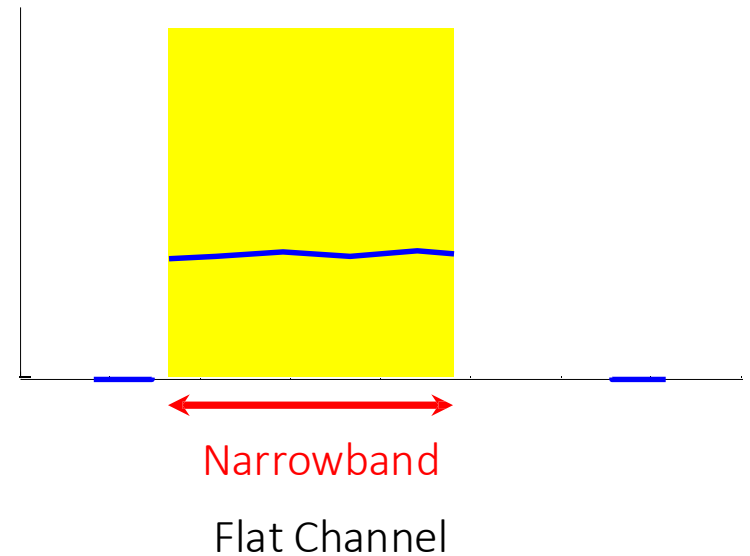
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



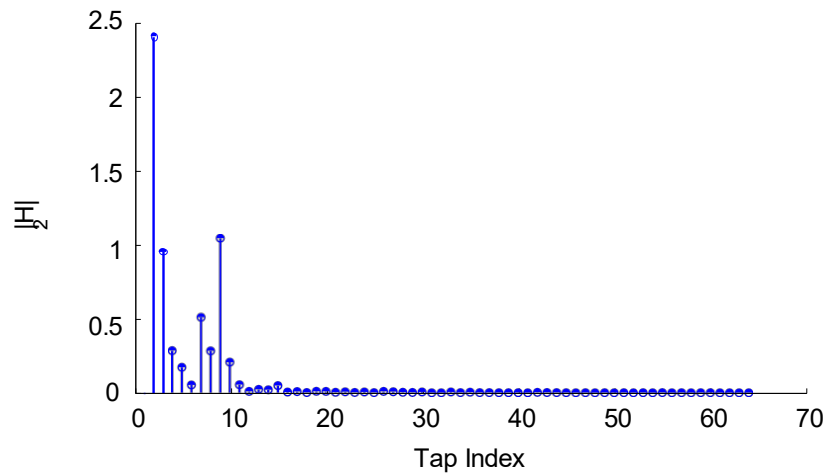
Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



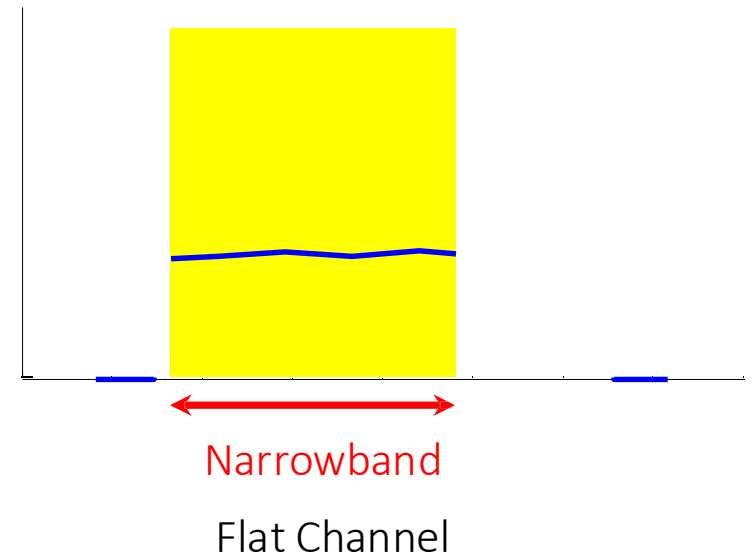
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



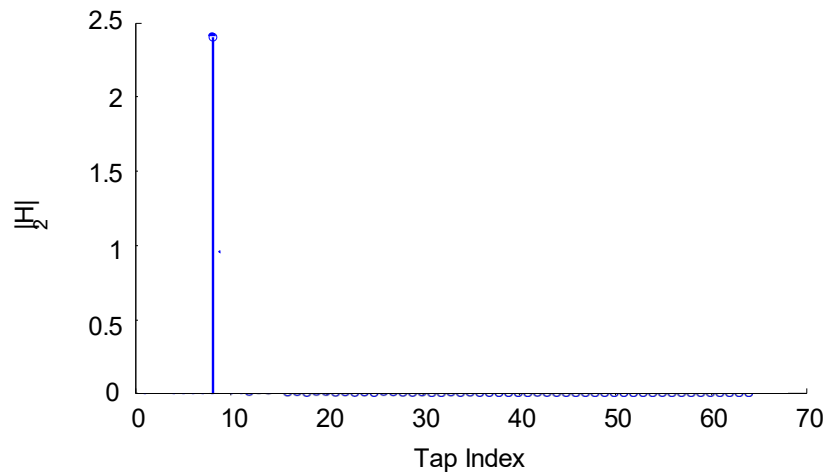
Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



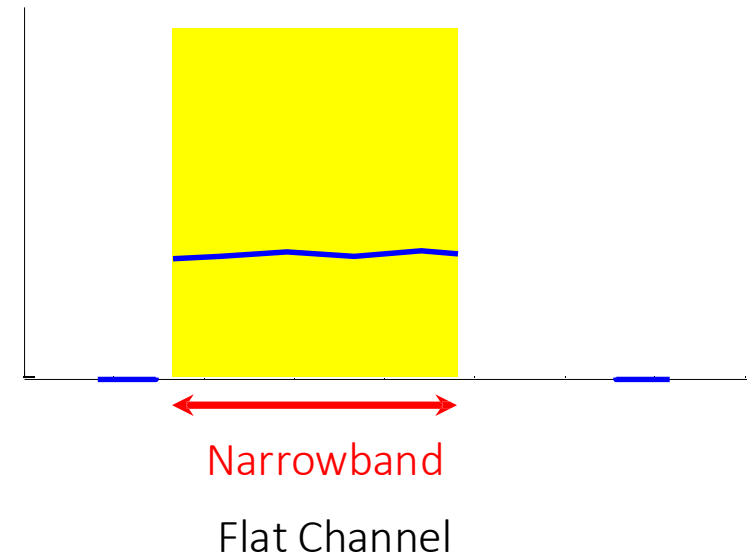
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



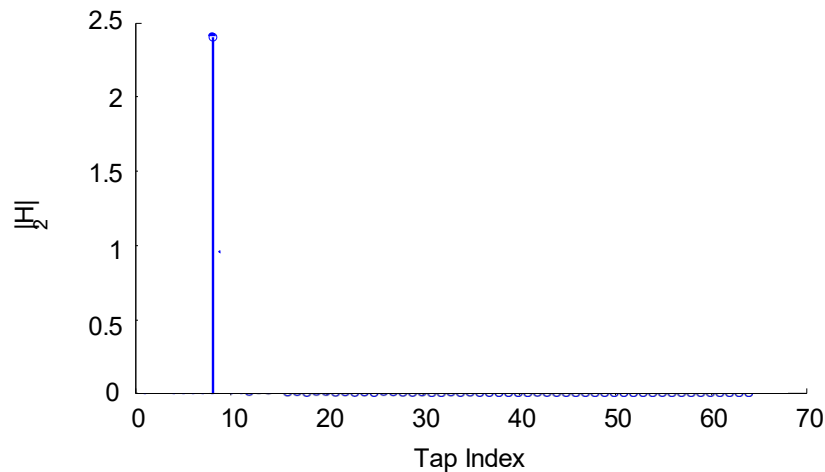
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = hx(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

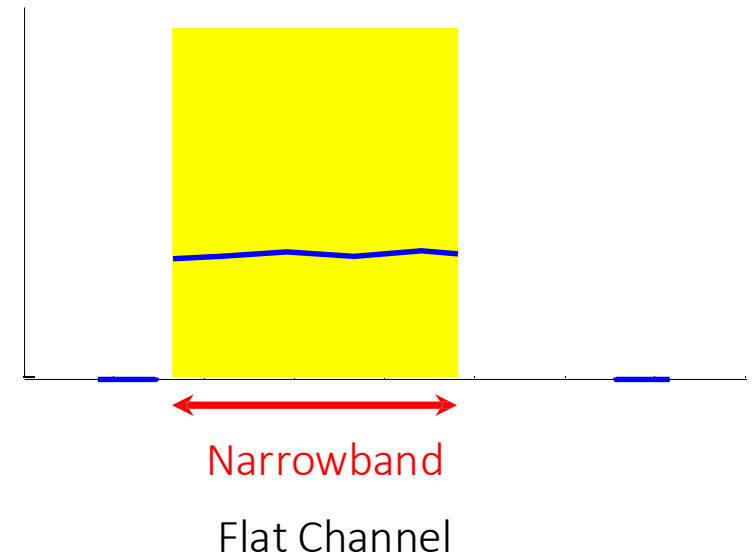
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx h x(t) \quad \Leftrightarrow \quad h X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



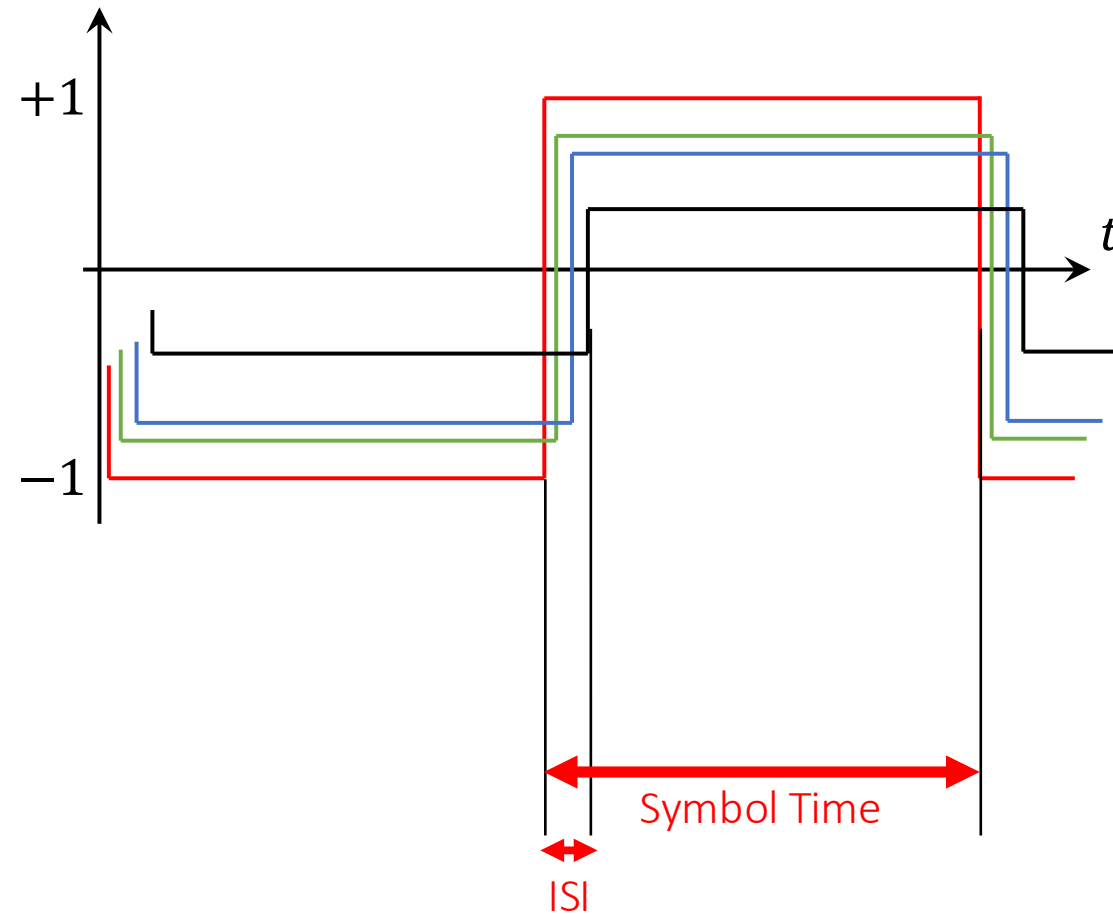
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = h x(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

Narrowband vs. Wideband Channel

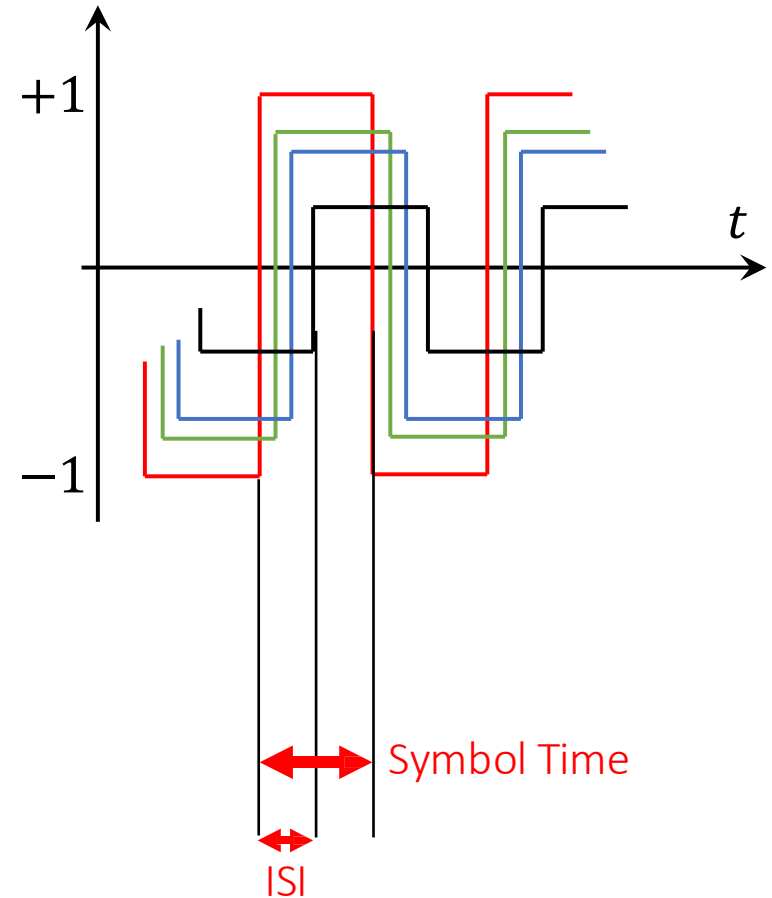
Narrowband Channel

$$h x(t)$$



Wideband Channel

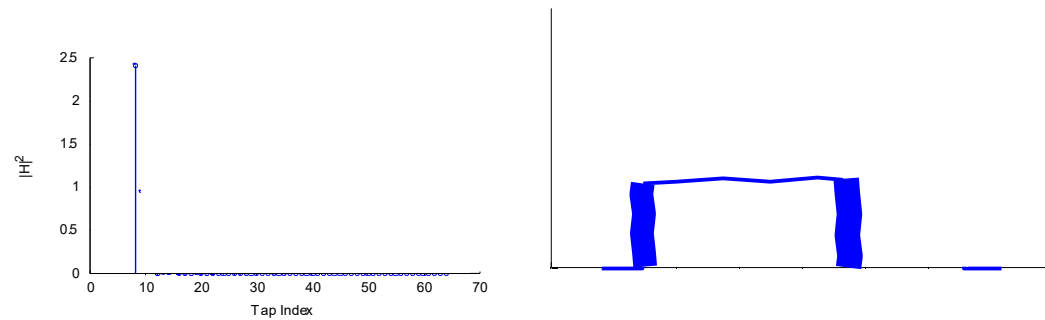
$$h(t) * x(t)$$



Narrowband vs. Wideband Channel

Narrowband Channel

$$h x(t) \iff h X(f)$$

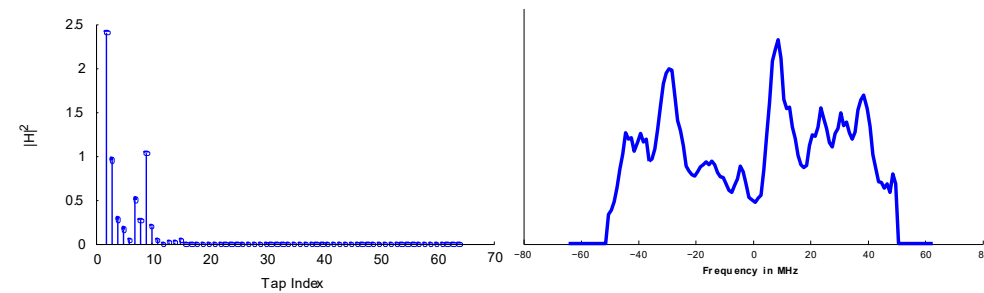


≈ Single Tap

≈ Flat Channel

Wideband Channel

$$h(t) * x(t) \iff H(f) X(f)$$



Multi-tap
Channel
(ISI)

Frequency
Selective Channel

Need to correct for ISI to
be able to decode
correctly!

Estimating and Correcting for multi-tap channel is hard
Simplify processing using: OFDM